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```
./gen > input
# python3 gen.py > input
./$CODE1 < input > $CODE1.out
./$CODE2 < input > $CODE2.out
cmp $CODE1.out $CODE2.out || break
done
# 多重解, ifstream in(argv[1]);
CODE="a"
set -e
g++ $CODE.cpp -o $CODE
g++ gen.cpp -o gen
g++ checker.cpp -o checker
for ((i=1;;i++))
do
./gen > input
./$CODE < input > $CODE.out
./checker $CODE.out < input || break
# 互動
CODE="a"
set -e
g++ $CODE.cpp -o $CODE
g++ checker.cpp -o checker
PIPE_IN="in"
PIPE_OUT="out"
trap 'rm -f $PIPE_IN $PIPE_OUT' EXIT
mkfifo $PIPE_IN $PIPE_OUT
for ((i=1;;i++))
do
echo "--- Testing: Case #${i} ---"
./$CODE < $PIPE_IN > $PIPE_OUT &
(exec 3>$PIPE_IN 4<$PIPE_OUT; ./checker <&4 >&3) || break
done
# 參考 checker
ll AC(int n) { return ans; }
void WA(string log = "") { cerr << log << endl; exit(1); }
void checkAC(string log = "") {
string trash;
if (cin >> trash) WA("redundant output\n" + log);
}
int main() {
int n = uniform_int_distribution<int>(1, 10)(rng);
ll sol = AC(n);
stringstream log;
cout << n << endl;
log << n << endl;
log << "judge: " << endl;
log << "team: " << endl;
WA(log.str());
checkAC();
return 0;
}
```

1.3 Int128 [eb8585]

```
using i128 = __int128_t; // 1.7E38
istream &operator>>(istream &is, i128 &a) {
i128 sgn = 1; a = 0;
string s; is >> s;
for (auto c : s) {
if (c == '-') sgn = -1;
else a = a * 10 + c - '0';
}
a *= sgn;
return is;
} // 7ea35c
ostream &operator<<(ostream &os, i128 a) {
if (a == 0) return os << 0;
string res;
if (a < 0) os << '-', a = -a;
while (a) res.push_back(a % 10 + '0'), a /= 10;
reverse(res.begin(), res.end());
os << res;
return os;
} // e5d7e9
```

1.4 Div [5a160f]

```
template<class T> T floor(T a, T b)
{ return a / b - (a % b != 0 && ((a < 0) != (b < 0))); }
template<class T> T ceil(T a, T b)
{ return a / b + (a % b != 0 && ((a < 0) == (b < 0))); }
// a / b < x -> floor(a, b) + 1 <= x
// a / b <= x -> ceil(a, b) <= x
// x < a / b -> x <= ceil(a, b) - 1
// x <= a / b -> x <= floor(a, b)
```

1.5 Pbds and Cmp [6e57d7]

```
// pbds_multiset:
// 必須 = , 其他 STL 都不能
// lower_bound, upper_bound 交換用法
// 不要用 find(x), erase(x), 用 upper_bound 拿到 it 再判斷
// order_of_key(x) 幾個元素嚴格小於 x
#include <ext/pb_ds/assoc_container.hpp>
#include <ext/pb_ds/tree_policy.hpp>
using namespace __gnu_pbds;
template<class T, class F = less<T>>
using pbds_set = tree<T, null_type,
, F, rb_tree_tag, tree_order_statistics_node_update>;
template<class T, class F = less_equal<T>>
```

1 Basic

1.1 Shell Scripts [54e75c]

```
# ~/.bashrc
CXX="g++
-std=c++17 -Wall -Wextra -g -fsanitize=address,undefined"
run() {
local b=./$(hash $1).out
[ -x $b ] || { CXX $1 -o $b && $b "$@"; }
}
hash() {
cpp $1 -dD -
P -fpreprocessed | tr -d "[:space:]" | md5sum | cut -c -6
}
```

1.2 Debug [33ccce]

```
# 對拍
CODE1="a"
CODE2="ac"
set -e
g++ $CODE1.cpp -o $CODE1
g++ $CODE2.cpp -o $CODE2
g++ gen.cpp -o gen
for ((i=1;;i++))
do
echo "--- Testing: Case #${i} ---"
```

```
using pbds_multiset = tree<T, null_type
    , F, rb_tree_tag, tree_order_statistics_node_update>;

// priority queue 小到大是 >, set 是 <
auto cmp = [](int i, int j) { return i > j; };
priority_queue<int, vector<int>, decltype(cmp)> pq(cmp);

vector<int> a{1, 2, 5, 4, 3}; // 小心不要改到 a
auto cmp = [&a](int i, int j) { return a[i] > a[j]; };
priority_queue<int, vector<int>, decltype(cmp)> pq(cmp);

vector<int> v{1, 2, 3, 4, 5};
upper_bound(v.begin(), v.end(), 2, [](int a, int b)
{ return a < b; }); // find first b that a < b, a is 2
lower_bound(v.begin(), v.end(), 2, [](int a, int b)
{ return a < b; }); // find first a that a < b fail, b is 2
```

2 Graph

2.1 Binary Lifting LCA [5438e3]

```
const int N = 2E5;
const int Lg = __lg(N); // __lg(max(N, Qi)), [0, Lg]
int up[N][Lg + 1];
vector<int> dep, dfn;
void build(int n, vector<vector<int>> &g, int rt = 0) {
    dep.assign(n, 0), dfn.assign(n, 0);
    int cur = 0;
    [&](this auto &&self, int x, int p) -> void {
        dfn[x] = cur++;
        up[x][0] = p;
        for (int i = 1; i <= Lg; i++) {
            int nxt = up[x][i - 1];
            up[x][i] = up[nxt][i - 1];
        }
        for (auto y : g[x]) {
            if (y == p) continue;
            up[y][0] = x;
            dep[y] = dep[x] + 1;
            self(y, x);
        }
    } (rt, rt);
}
int lca(int a, int b) {
    if (dep[a] < dep[b]) swap(a, b);
    int pull = dep[a] - dep[b];
    for (int i = 0; i <= Lg; i++)
        if (pull >> i & 1) a = up[a][i];
    if (a == b) return a;
    for (int i = Lg; i >= 0; i--)
        if (up[a][i] != up[b][i])
            a = up[a][i], b = up[b][i];
    return up[a][0];
}
int jump(int x, int k) {
    for (int i = Lg; i >= 0; i--)
        if (k >> i & 1) x = up[x][i];
    return x;
}
int dist(int a, int b) {
    return dep[a] + dep[b] - 2 * dep[lca(a, b)];
}
```

2.2 Heavy Light Decomposition [0db4d5]

```
struct HLD {
    int n, cur = 0;
    vector<int> siz, top, dep, par, in, out, seq;
    vector<vector<int>> adj;
    HLD(int n) : n(n), siz(n),
        top(n), dep(n), par(n), in(n), out(n), seq(n), adj(n) {}
    void addEdge(int u, int v) {
        adj[u].push_back(v);
        adj[v].push_back(u);
    }
    void work(int rt = 0) {
        top[rt] = rt;
        dep[rt] = 0;
        par[rt] = -1;
        dfs1(rt); dfs2(rt);
    }
    void dfs1(int u) {
        if (par[u] != -1) adj
            [u].erase(find(adj[u].begin(), adj[u].end(), par[u]));
        siz[u] = 1;
        for (auto &v : adj[u]) {
            par[v] = u, dep[v] = dep[u] + 1;
            dfs1(v);
            siz[u] += siz[v];
            if (siz[v] > siz[adj[u][0]]) {
                swap(v, adj[u][0]);
            } // 讓 adj[u][0] 是重子節點
        }
    }
    void dfs2(int u) {
        in[u] = cur++;
        seq[in[u]] = u; // dfn 對應的編號
        for (auto v : adj[u]) {
            top[v] = v == adj[u][0] ? top[u] : v;
            dfs2(v);
        }
    }
}
```

```
    out[u] = cur;
}
int lca(int u, int v) {
    while (top[u] != top[v]) {
        if (dep[top[u]] > dep[top[v]]) {
            u = par[top[u]];
        } else {
            v = par[top[v]];
        }
    }
    return dep[u] < dep[v] ? u : v;
}
int dist(int u, int v) {
    return dep[u] + dep[v] - 2 * dep[lca(u, v)];
}
int jump(int u, int k) {
    if (dep[u] < k) return -1;
    int d = dep[u] - k;
    while (dep[top[u]] > d) u = par[top[u]];
    return seq[in[u] - dep[u] + d];
} // c26ac4
bool isAncestor(int u, int v) {
    return in[u] <= in[v] && in[v] < out[u];
} // 2ca857
int rootedParent(int rt, int v) {
    if (rt == v) return rt;
    if (!isAncestor(v, rt)) return par[v];
    auto it = upper_bound(adj[v].begin(), adj[v].end(), rt,
        [&](int x, int y) {
            return in[x] < in[y];
        }) - 1;
    return *it;
} // c8db9f
int rootedSize(int rt, int v) {
    if (rt == v) return n;
    if (!isAncestor(v, rt)) return siz[v];
    return n - siz[rootedParent(rt, v)];
} // d7dea9
int rootedLca(int rt, int a, int b) {
    return lca(rt, a) ^ lca(a, b) ^ lca(b, rt);
} // da7f79
};
```

2.3 Centroid Decomposition [7ef37f]

```
vector<bool> vis(n);
vector<int> siz(n), par(n, -1);
auto findSize = [&](auto &&self, int u, int p) -> int {
    siz[u] = 1;
    for (int v : g[u]) {
        if (v == p || vis[v]) continue;
        siz[u] += self(self, v, u);
    }
    return siz[u];
};
auto findCen = [&](auto &&self, int u, int p, int sz) -> int {
    for (int v : g[u]) {
        if (v == p || vis[v]) continue;
        if (siz[v] * 2 > sz) return self(self, v, u, sz);
    }
    return u;
};
auto build = [&](auto &&self, int u, int p) -> void {
    findSize(findSize, u, p);
    int c = findCen(findCen, u, -1, siz[u]);
    vis[c] = true, par[c] = p;
    for (int v : g[c]) if (!vis[v]) self(self, v, c);
}; build(build, 0, -1);
```

2.4 SCC [414637]

```
struct SCC {
    int n, cur = 0, cnt = 0;
    vector<vector<int>> adj;
    vector<int> stk, dfn, low, bel;
    SCC(int n) : n(n), adj(n), dfn(n, -1), low(n), bel(n, -1) {}
    void addEdge(int u, int v) { adj[u].push_back(v); }
    void dfs(int x) {
        dfn[x] = low[x] = cur++;
        stk.push_back(x);
        for (auto y : adj[x]) {
            if (dfn[y] == -1) {
                dfs(y);
                low[x] = min(low[x], low[y]);
            } else if (bel[y] == -1) {
                low[x] = min(low[x], dfn[y]);
            }
        }
        if (dfn[x] == low[x]) {
            int y;
            do {
                y = stk.back();
                bel[y] = cnt;
                stk.pop_back();
            } while (y != x);
            cnt++;
        }
    }
    vector<int> work() {
        for (int i = 0; i < n; i++)
            if (dfn[i] == -1) dfs(i);
        return bel;
    }
}
```

```

} // 1ebd5a
struct Graph {
    int n;
    vector<pair<int, int>> edges;
    vector<int> siz, cnte;
};
Graph compress() {
    Graph g;
    g.n = cnt;
    g.siz.resize(cnt), g.cnte.resize(cnt);
    for (int i = 0; i < n; i++) {
        g.siz[bel[i]]++;
        for (auto j : adj[i]) {
            if (bel[i] != bel[j]) {
                g.edges.emplace_back(bel[i], bel[j]);
            } else {
                g.cnte[bel[i]]++;
            }
        }
    }
    return g;
}
};

```

2.5 2-SAT [093413]

```

struct TwoSat : SCC {
    vector<int> ans;
    TwoSat(int n) : SCC(2 * n), ans(n) {}
    void addClause(int u, bool a, int v, bool b) {
        addEdge(2 * u + !a, 2 * v + b);
        addEdge(2 * v + !b, 2 * u + a);
    }
    void ifThen(int u, bool a, int v, bool b) {
        // 必取 A: not A -> A
        addEdge(2 * u + a, 2 * v + b);
    }
    bool work() {
        SCC::work(); // bel[!a] > bel[a] => !a -> a
        for (int i = 0; i < n / 2; i++) {
            if (bel[2 * i] == bel[2 * i + 1]) return false;
            ans[i] = bel[2 * i] > bel[2 * i + 1];
        }
        return true;
    }
}
};

```

2.6 EBCC [a84a20]

```

struct EBCC { // CF/contest/1986/pF
    int n, cur = 0, cnt = 0;
    vector<vector<int>> adj;
    vector<int> stk, dfn, low, bel;
    vector<pair<int, int>> bridges; // 關鍵邊
    EBCC(int n) : n(n), adj(n), low(n), dfn(n, -1), bel(n, -1) {}
    void addEdge(int u, int v) {
        adj[u].push_back(v);
        adj[v].push_back(u);
    }
    void dfs(int x, int p) {
        dfn[x] = low[x] = cur++;
        stk.push_back(x);
        for (auto y : adj[x]) {
            if (y == p) continue;
            if (dfn[y] == -1) {
                dfs(y, x);
                low[x] = min(low[x], low[y]);
                if (low[y] > dfn[x]) {
                    bridges.emplace_back(x, y);
                }
            } else if (bel[y] == -1) {
                low[x] = min(low[x], dfn[y]);
            }
        }
        if (dfn[x] == low[x]) {
            int y;
            do {
                y = stk.back();
                bel[y] = cnt;
                stk.pop_back();
            } while (y != x);
            cnt++;
        }
    }
    vector<int> work() { // not connected
        for (int i = 0; i < n; i++)
            if (dfn[i] == -1) dfs(i, -1);
        return bel;
    } // e0899d
    struct Graph {
        int n;
        vector<pair<int, int>> edges;
        vector<int> siz, cnte;
    };
    Graph compress() {
        Graph g; g.n = cnt;
        g.siz.resize(cnt); g.cnte.resize(cnt);
        for (int i = 0; i < n; i++) {
            g.siz[bel[i]]++;
            for (auto j : adj[i]) {
                if (bel[i] < bel[j]) {
                    g.edges.emplace_back(bel[i], bel[j]);
                }
            }
        }
    }
};

```

```

        } else if (bel[i] == bel[j] && i < j) {
            g.cnte[bel[i]]++;
        }
    }
    return g;
}
};

```

2.7 VBCC [30c4d8]

```

struct VBCC {
    int n, cur = 0, cnt = 0;
    vector<vector<int>> adj, bcc;
    vector<int> dfn, low, stk;
    vector<bool> ap;
    VBCC(int n) : n(n), adj(n), dfn(n, 0), low(n, 0), ap(n) {}
    void addEdge(int u, int v) {
        adj[u].push_back(v);
        adj[v].push_back(u);
    }
    void dfs(int x, int p = -1) {
        int ch = 0;
        dfn[x] = low[x] = ++cur;
        stk.push_back(x);
        for (int y : adj[x]) {
            if (!dfn[y]) {
                dfs(y, x);
                ch++;
                low[x] = min(low[x], low[y]);
                if (dfn[x] <= low[y]) {
                    ap[x] = true;
                    bcc.emplace_back();
                    int v;
                    do {
                        v = stk.back();
                        stk.pop_back();
                        bcc.back().push_back(v);
                    } while (v != y);
                    bcc.back().push_back(x);
                    cnt++;
                }
            } else if (dfn[y] < dfn[x] && y != p)
                low[x] = min(low[x], dfn[y]);
        }
        if (p == -1 && ch < 2) ap[x] = false;
    }
    void work() {
        for (int i = 0; i < n; i++)
            if (!dfn[i])
                if (adj[i].empty()) {
                    bcc.emplace_back(1, i);
                    cnt++;
                } else dfs(i);
    } // 674cc5
    struct Graph {
        int n;
        vector<pair<int, int>> edges;
        vector<int> bel, siz, cnte;
    };
    Graph compress() {
        Graph g;
        g.bel.assign(n, -1);
        g.siz.assign(cnt, 0);
        g.cnte.assign(cnt, 0);
        g.n = cnt;
        for (int i = 0; i < n; i++) {
            if (ap[i]) {
                g.bel[i] = g.n++;
                g.siz.push_back(1);
                g.cnte.push_back(0);
            }
        }
        vector<bool> in(n);
        for (int i = 0; i < cnt; i++) {
            for (int u : bcc[i]) in[u] = true;
            int cnte = 0;
            for (int u : bcc[i]) {
                if (ap[u]) {
                    g.edges.emplace_back(g.bel[u], i);
                } else {
                    g.bel[u] = i, g.siz[i]++;
                }
            }
            for (int v : adj[u]) if (in[v]) cnte++;
        }
        g.cnte[i] = cnte / 2;
        for (int u : bcc[i]) in[u] = false;
    }
    return g;
}
};

```

2.8 Functional Graph [060d10]

```

const int N = 2E5;
const int Lg = __lg(N); // __lg(max(n, qi)), [0, Lg]
int cht[N][Lg];
struct FunctionalGraph {
    int n, cnt = 0;
    vector<int> g, bel, id, cycsz, in, top, hei;
    FunctionalGraph(const vector<int> &g) : n(g.size())
        , g(g), bel(n, -1), id(n), in(n), top(n, -1), hei(n) {}
};

```

```

for (int i = 0; i < n; i++)
    cht[i][0] = g[i], in[g[i]]++;
for (int i = 1; i <= Lg; i++)
    for (int u = 0; u < n; u++) {
        int nxt = cht[u][i - 1];
        cht[u][i] = cht[nxt][i - 1];
    }
for (int i = 0; i < n; i++)
    if (in[i] == 0) label(i);
for (int i = 0; i < n; i++)
    if (top[i] == -1) label(i);
}
void label(int u) {
    vector<int> p; int cur = u;
    while (top[cur] == -1) {
        top[cur] = u;
        p.push_back(cur);
        cur = g[cur];
    }
    auto s = find(p.begin(), p.end(), cur);
    vector<int> cyc(s, p.end());
    p.erase(s, p.end()); p.push_back(cur);
    for (int i = 0; i < (int)cyc.size(); i++)
        bel[cyc[i]] = cnt, id[cyc[i]] = i, hei[cyc[i]] = cyc.size();
    if (!cyc.empty())
        ++cnt, cycsz.push_back(cyc.size());
    for (int i = p.size() - 1; i > 0; i--)
        id[p[i] - 1] = id[p[i]] - 1, hei[p[i] - 1] = hei[p[i]] + 1;
}
int jump(int u, int k) {
    for (int b = 0; k > 0; b++) {
        if (k & 1) u = cht[u][b];
        k >>= 1;
    }
    return u;
}
};

```

2.9 Euler [13ce48]

```

vector<int> ans;
[&](this auto &&self, int u) -> void {
    while (g[u].size()) {
        int v = *g[u].begin();
        g[u].erase(v);
        self(v);
    }
    ans.push_back(u);
} (0); reverse(ans.begin(), ans.end());

```

2.10 Virtual Tree [b2b0eb]

// 多次詢問給某些關鍵點，虛樹可達成快速樹 DP (前處理每個點)
 // 例如這題是有權樹，給一些關鍵點，求跟 vertex 1 隔開的最小成本
 // 前處理 root 到所有點的最小邊權

```

void solve(int n, vector<vector<pair<int, int>>> &g, int q) {
    build(n, g); // build LCA
    vector<int> dis(n, 1E9); // root 到各點的最小邊權
    [&](this auto &&self, int x, int p) -> void {
        for (auto [y, w] : g[x]) {
            if (y == p) continue;
            dis[y] = min(w, dis[x]);
            self(y, x);
        }
    } (0, -1);
    vector<vector<int>> vt(n);
    vector<int> isKey(n);
    vector<ll> dp(n);
    while (q--) {
        int m; cin >> m;
        vector<int> key(m);
        for (int i = 0; i < m; i++) {
            cin >> key[i];
            isKey[i - key[i]] = 1;
        }
        key.push_back(0); // 固定 0 為 root，看題目需求
        sort(key.begin(), key.end(), [&](int a, int b) {
            return dfn[a] < dfn[b];
        }); // 要 sort 再 insert
        vector<int> stk; // 維護一條虛樹鏈
        for (int k : key) {
            if (stk.empty()) {
                stk.push_back(k);
                continue;
            }
            int l = lca(stk.back(), k);
            if (l == stk.back()) { // 直接一條鏈
                stk.push_back(k);
                continue;
            } // 次頂是 lca 的子孫，直接跟最頂連邊
            while (stk.size() > 1 && dfn[stk[stk.size() - 2]] > dfn[l]) {
                vt[stk[stk.size() - 2]].push_back(stk.back());
                stk.pop_back();
            } // 保證 back 是 lca 的子孫，且次頂不是 lca 的子孫
            vt[l].push_back(stk.back());
            if (stk.size() < 2 || stk[stk.size() - 2] != l) stk.back() = l;
        }
    }
}

```

```

else stk
    .pop_back(); // lca 的子孫已經用不到了，直接換成 lca
stk.push_back(k);
}
while (stk.size() > 1) {
    vt[stk[stk.size() - 2]].push_back(stk.back());
    stk.pop_back();
}
int rt = stk[0]; stk.clear();
[&](this auto &&self, int x) -> void {
    for (auto y : vt[x]) {
        self(y);
        if (isKey[y]) { // 直接砍了
            dp[x] += dis[y];
        } else { // 不砍 or 砍
            dp[x] += min<ll>(dp[y], dis[y]);
        } // 記得 reset
        isKey[y] = dp[y] = 0;
    }
    vt[x].clear(); // 記得 reset
} (rt);
cout << dp[rt] << "\n";
isKey[rt] = 0;
dp[rt] = 0; // 最後 reset root
}
}

```

2.11 Dominator Tree [a54c17]

// dom
 存起點到達此點的必經的上個節點(起點 = 自己)，無法到達 = -1

```

struct DominatorTree {
    int n, id = 0;
    vector<vector<int>> adj, radj, bucket;
    vector<int> sdом, dom, vis, rev, pa, rt, mn, res;
    DominatorTree(int
        n) : n(n), adj(n), radj(n), bucket(n), sdом(n), dom(n,
            -1), vis(n, -1), rev(n), pa(n), rt(n), mn(n), res(n) {}
    void addEdge(int u, int v) { adj[u].push_back(v); }
    int query(int v, int x) {
        if (rt[v] == v) return x ? -1 : v;
        int p = query(rt[v], 1);
        if (p == -1) return x ? rt[v] : mn[v];
        if (sdом[mn[v]] > sdом[mn[rt[v]]]) mn[v] = mn[rt[v]];
        rt[v] = p;
        return x ? p : mn[v];
    }
    void dfs(int v) {
        vis[v] = id, rev[id] = v;
        rt[id] = mn[id] = sdом[id] = id, id++;
        for (int u : adj[v]) {
            if (vis[u] == -1) dfs(u), pa[vis[u]] = vis[v];
            radj[vis[u]].push_back(vis[v]);
        }
    }
    vector<int> build(int s) {
        dfs(s);
        for (int i = id - 1; i >= 0; i--) {
            for (int u : radj[i])
                sdом[i] = min(sdом[i], sdом[query(u, 0)]);
            if (i) bucket[sdом[i]].push_back(i);
            for (int u : bucket[i]) {
                int p = query(u, 0);
                dom[u] = sdом[p] == i ? i : p;
            }
            if (i) rt[i] = pa[i];
        }
        res.assign(n, -1);
        for (int i = 1; i < id; i++)
            if (dom[i] != sdом[i]) dom[i] = dom[dom[i]];
        for (int i = 1; i < id; i++) res[rev[i]] = rev[dom[i]];
        res[s] = s;
        for (int i = 0; i < n; i++) dom[i] = res[i];
        return dom;
    }
}
};

```

2.12 Theorems

2.12.1 Euler's Theorem

- 無向圖是歐拉圖：
 - 非零度頂點是連通的
 - 頂點的度數都是偶數
- 無向圖是半歐拉圖(有路沒有環)：
 - 非零度頂點是連通的
 - 恰有 2 個奇度頂點
- 有向圖是歐拉圖：
 - 非零度頂點是強連通的
 - 每個頂點的入度和出度相等
- 有向圖是半歐拉圖(有路沒有環)：
 - 非零度頂點是弱連通的
 - 至多一個頂點的出度與入度之差為 1
 - 至多一個頂點的入度與出度之差為 1
 - 其他頂點的入度和出度相等

2.12.2 Diff Constraints

限制	邊
$x_v - x_u \leq c$	$u \rightarrow v, c$
$x_v - x_u \geq c$	$v \rightarrow u, -c$
$x_v - x_u = c$	上面兩條都建
$x_v \leq c$	$Z \rightarrow v, c$ (Z 為參考點)
$x_v \geq c$	$v \rightarrow Z, -c$

- 用到參考點 Z 時，答案取 $x_i = d_i - d_Z$
- 初始化 $d_i = 0$ (等同超級源點連 0 邊)，有解 $\iff$ 無負環
- $\max(x_t - x_s) = \text{dist}_{\min}(s \rightarrow t)$; t 不可達 $\Rightarrow$ 無界
- 取 $\min \Rightarrow$ 寫成 $\geq$ 形式跑最長路

3 Data Structure

3.1 Fenwick [d41d8c]

```
template<class T> struct Fenwick {
    int n;
    vector<T> a;
    Fenwick(int n) : n(n), a(n) {}
    void add(int x, const T &v) {
        for (x++; x <= n; x += x & -x) a[x - 1] = a[x - 1] + v;
    }
    T sum(int x) {
        T v{};
        for (; x > 0; x -= x & -x) v = v + a[x - 1];
        return v;
    }
    T rangeSum(int l, int r) { return sum(r) - sum(l); } /*
    int select(const T &k, int start = 0) {
        // 找到最小的 x, 使得 sum(x + 1) - sum(start) > k
        // prefix sum 要有單調性
        int x = 0;
        T cur = -sum(start);
        for (int i = 1 << __lg(n); i > 0; i /= 2)
            if (x + i <= n && cur + a[x + i - 1] <= k)
                x += i, cur = cur + a[x - 1];
        return x;
    } */ // 8e1de2
}; // 9083e4

template<class T> struct Fenwick2D {
    int n, m;
    vector<vector<T>> a;
    Fenwick2D(int n, int m) : n(n), m(m), a(n, vector<T>(m)) {}
    void add(int x, int y, const T &v) {
        for (int i = x + 1; i <= n; i += i & -i)
            for (int j = y + 1; j <= m; j += j & -j)
                a[i - 1][j - 1] = a[i - 1][j - 1] + v;
    }
    T sum(int x, int y) {
        T v{};
        for (int i = x; i > 0; i -= i & -i)
            for (int j = y; j > 0; j -= j & -j)
                v = v + a[i - 1][j - 1];
        return v;
    }
    T rangeSum(int x1, int y1, int x2, int y2) {
        return sum(x2, y2) - sum(x1, y2) - sum(x2, y1) + sum(x1, y1);
    }
}; // a71e39

template<class T> struct RangeFenwick {
    int n;
    vector<T> d, di;
    RangeFenwick(int n) : n(n), d(n), di(n) {}
    void add(int x, const T &v) {
        T vi = v * (x + 1);
        for (int i = x + 1; i <= n; i += i & -i)
            d[i - 1] = d[i - 1] + v, di[i - 1] = di[i - 1] + vi;
    }
    void rangeAdd(int l, int r, const T &v) { add(l, v), add(r, -v); }
    T sum(int x) {
        T v{};
        for (int i = x; i > 0; i -= i & -i)
            v = v + T(x + 1) * d[i - 1] - di[i - 1];
        return v;
    }
    T rangeSum(int l, int r) { return sum(r) - sum(l); } /*
    int select(const T &k, int start = 0) {
        int x = 0;
        T cur{}, curi{}, sub = sum(start);
        for (int i = 1 << __lg(n); i; i /= 2) {
            if (x + i <= n && T(x + i + 1) * (cur +
                d[x + i - 1]) - (curi + di[x + i - 1]) - sub <= k) {
                x += i;
                cur = cur + d[x - 1], curi = curi + di[x - 1];
            }
        }
        return x;
    } */ // e93ec7
}; // 03784e

template<class T> struct RangeFenwick2D {
    int n, m;
    vector<vector<T>> d, di, dj, dij;
    RangeFenwick2D(int n, int m) : n(n), m(m), dij(n, vector<T>(m)) {}
    { d = di = dj = dij; }
    void add(int x, int y, const T &v) {
        T vi = v * (x + 1), vj = v * (y + 1);
```

```
        T vij = v * (x + 1) * (y + 1);
        for (int i = x + 1; i <= n; i += i & -i)
            for (int j = y + 1; j <= m; j += j & -j) {
                d[i - 1][j - 1] = d[i - 1][j - 1] + v;
                di[i - 1][j - 1] = di[i - 1][j - 1] + vi;
                dj[i - 1][j - 1] = dj[i - 1][j - 1] + vj;
                dij[i - 1][j - 1] = dij[i - 1][j - 1] + vij;
            }
    }
    void rangeAdd(int x1, int y1, int x2, int y2, const T &v) {
        add(x1, y1, v), add(x2, y2, v);
        add(x1, y2, -v), add(x2, y1, -v);
    }
    T sum(int x, int y) {
        T v{};
        for (int i = x; i > 0; i -= i & -i)
            for (int j = y; j > 0; j -= j & -j)
                v = v + (x + 1) * (y + 1) * d[i - 1][j - 1]
                    - (y + 1) * di[i - 1][j - 1]
                    - (x + 1) * dj[i - 1][j - 1]
                    + dij[i - 1][j - 1];
        return v;
    }
    T rangeSum(int x1, int y1, int x2, int y2) {
        return sum(x2, y2) - sum(x1, y2) - sum(x2, y1) + sum(x1, y1);
    }
}; // 60ea46
```

3.2 Lazy Segment Tree [d41d8c]

```
template<class Info, class Tag> struct LazySegmentTree {
    int n;
    vector<Info> info;
    vector<Tag> tag;
    LazySegmentTree(int n) : n(n), info(4 << __lg(n)), tag(4 << __lg(n)) {}
    LazySegmentTree(const vector<Info> &v) : n(v.size()), info(4 << __lg(n)), tag(4 << __lg(n)) {}
    auto build = [&](auto &&self, int p, int l, int r) {
        if (r - l == 1) {
            info[p] = v[l];
            return;
        }
        int m = (l + r) / 2;
        self(self, 2 * p, l, m);
        self(self, 2 * p + 1, m, r);
        pull(p);
    };
    build(build, 1, 0, n);
    void pull(int p) { info[p] = info[2 * p] + info[2 * p + 1]; }
    void apply(int p, int l, int r, const Tag &t) {
        info[p].apply(t, l, r);
        tag[p].apply(t);
    }
    void push(int p, int l, int r) {
        int m = (l + r) / 2;
        apply(2 * p, l, m, tag[p]);
        apply(2 * p + 1, m, r, tag[p]);
        tag[p] = Tag{};
    }
    void modify(int x, const Info &i) { modify(1, 0, n, x, i); }
    void modify(int p, int l, int r, int x, const Info &i) {
        if (r - l == 1) {
            info[p] = i; // add use +=
            return;
        }
        int m = (l + r) / 2;
        push(p, l, r);
        if (x < m) modify(2 * p, l, m, x, i);
        else modify(2 * p + 1, m, r, x, i);
        pull(p);
    }
    Info query(int ql, int qr) { return query(1, 0, n, ql, qr); }
    Info query(int p, int l, int r, int ql, int qr) {
        if (qr <= l || ql >= r) return Info{};
        if (ql <= l && r <= qr) return info[p];
        int m = (l + r) / 2;
        push(p, l, r);
        return query(2 * p, l, m, ql, qr) + query(2 * p + 1, m, r, ql, qr);
    }
    void rangeApply(int ql, int qr, const Tag &t) { rangeApply(1, 0, n, ql, qr, t); }
    void rangeApply(int p, int l, int r, int ql, int qr, const Tag &t) {
        if (qr <= l || ql >= r) return;
        if (ql <= l && r <= qr) {
            apply(p, l, r, t.offset(l - ql)); // tag 可位移 (l - ql)
            return;
        }
        int m = (l + r) / 2;
        push(p, l, r);
        rangeApply(2 * p, l, m, ql, qr, t);
        rangeApply(2 * p + 1, m, r, ql, qr, t);
        pull(p);
    }
    /*
    template<class F> int findFirst(int ql, int qr, F &&f) {
        return findFirst(1, 0, n, ql, qr, f);
    } // 若要找 last, 先右子樹遞迴即可
    */
```

```

template<class F> int
    findFirst(int p, int l, int r, int ql, int qr, F &&f) {
    if (qr <= l || ql >= r || !f(info[p])) return -1;
    if (r - l == 1) return l;
    int m = (l + r) / 2;
    push(p, l, r);
    int res = findFirst(2 * p, l, m, ql, qr, f);
    if (res == -1) res = findFirst(2 * p + 1, m, r, ql, qr, f);
    return res;
} /*
};
struct Tag { // 有些 Tag 不用 push 例如 sweepLine
    void apply(const Tag &t) {}
    Tag offset(int d) const { return *this; }
};
struct Info {
    void apply(const Tag &t, int l, int r) {} /*
    Info &operator=(const Info &i) { // partial assignment
        return *this;
    }
    Info &operator+=(const Info &i) { // leaf add
        return *this = *this + i;
    } /*
};
Info operator+(const Info &a, const Info &b) {
    Info c;
    return c;
}

```

3.3 Dynamic Segment Tree [d41d8c]

```

template<class Info> struct SegmentTreeDynamic {
    struct Node {
        Info info = Info();
        int lc = 0, rc = 0;
    };
    int n;
    vector<Node> nd;
    SegmentTreeDynamic(int n) : n(n), nd(1) {}
    int generate() {
        nd.emplace_back();
        return nd.size() - 1;
    }
    void pull(int p) {
        nd[p].info = nd[nd[p].lc].info + nd[nd[p].rc].info;
    }
    int modify(int x, const Info &i, int rt = 0) { return modify(rt, 0, n, x, i); }
    int modify(int p, int l, int r, int x, const Info &i) {
        if (p == 0) p = generate();
        if (r - l == 1) {
            nd[p].info = i; // add use +=
            return p;
        }
        int m = (l + r) / 2;
        if (x < m) nd[p].lc = modify(nd[p].lc, l, m, x, i);
        else nd[p].rc = modify(nd[p].rc, m, r, x, i);
        pull(p);
        return p;
    }
    int merge(int p, int q) { return merge(p, q, 0, n); }
    int merge(int p, int q, int l, int r) {
        if (p == 0 || q == 0) return p + q;
        if (r - l == 1) {
            nd[p].info += nd[q].info; // leaf add
            return p;
        }
        int m = (l + r) / 2;
        nd[p].lc = merge(nd[p].lc, nd[q].lc, l, m);
        nd[p].rc = merge(nd[p].rc, nd[q].rc, m, r);
        pull(p);
        return p;
    }
    Info query(int ql, int qr, int rt = 0) { return query(rt, 0, n, ql, qr); }
    Info query(int p, int l, int r, int ql, int qr) {
        if (l >= qr || r <= ql || p == 0) return Info();
        if (ql <= l && r <= qr) return nd[p].info;
        int m = (l + r) / 2;
        return query(nd[p].lc, l, m, ql, qr) + query(nd[p].rc, m, r, ql, qr);
    } /*
    template<class F> int findFirst(int ql, int qr, F &&f, int rt = 0) {
        return findFirst(rt, 0, n, ql, qr, f);
    }
    template<class F> int
        findFirst(int p, int l, int r, int ql, int qr, F &&f) {
        if (l >= qr || r <= ql || p == 0 || !f(nd[p].info)) return -1;
        if (r - l == 1) return l;
        int m = (l + r) / 2;
        int res = findFirst(nd[p].lc, l, m, ql, qr, f);
        if (res != -1) return res;
        return findFirst(nd[p].rc, m, r, ql, qr, f);
    } /*
};

```

3.4 Persistent Segment Tree [d41d8c]

```

template<class Info> struct PST {
    struct Node {
        Info info = Info();

```

```

        int lc = 0, rc = 0;
    };
    int n;
    vector<Node> nd;
    vector<int> rt;
    PST(int n) : n(n), nd(1), rt(1) {}
    PST(const vector<Info> &v) : n(v.size()), nd(1) {
        auto build = [&](auto &&self, int l, int r) -> int {
            int p = copy(0);
            if (r - l == 1) {
                nd[p].info = v[l];
                return p;
            }
            int m = (l + r) / 2;
            int lc = self(self, l, m);
            int rc = self(self, m, r);
            nd[p].lc = lc;
            nd[p].rc = rc;
            pull(nd[p]);
            return p;
        };
        rt.push_back(build(build, 0, n));
    }
    void pull(Node &p) { p.info = nd[p.lc].info + nd[p.rc].info; }
    int copy(int p) {
        nd.push_back(nd[p]);
        return nd.size() - 1;
    }
    void modify(int x, const Info &i, int ver = 0) {
        assert(rt.size() > ver);
        rt[ver] = modify(rt[ver], 0, n, x, i);
    }
    int modify(int p, int l, int r, int x, const Info &i) {
        p = copy(p);
        if (r - l == 1) {
            nd[p].info = i;
            return p;
        }
        int m = (l + r) / 2;
        if (x < m) nd[p].lc = modify(nd[p].lc, l, m, x, i);
        else nd[p].rc = modify(nd[p].rc, m, r, x, i);
        pull(nd[p]);
        return p;
    }
    Info query(int ql, int qr, int ver = 0) {
        return query(rt[ver], 0, n, ql, qr);
    }
    Info query(int p, int l, int r, int ql, int qr) {
        if (l >= qr || r <= ql || p == 0) return Info();
        if (ql <= l && r <= qr) return nd[p].info;
        int m = (l + r) / 2;
        return query(nd[p].lc, l, m, ql, qr) + query(nd[p].rc, m, r, ql, qr);
    }
    int createVersion(int ver) {
        rt.push_back(rt[ver]);
        return rt.size() - 1;
    }
};

```

3.5 Kth-element [d41d8c]

```

template<class T> struct StaticKth : PST<int> {
    int dct(T x) {
        return lower_bound(s.begin(), s.end(), x) - s.begin();
    }
    vector<T> a, s; // array, sorted
    StaticKth(const vector<T> &a, const vector<T> &s) : a(a), s(s), PST<int>(s.size()) {
        assert(is_sorted(s.begin(), s.end()));
        for (int i = 0; i < a.size(); i++) {
            createVersion(i);
            add(dct(a[i]), 1, i + 1);
        }
    }
    int work(int a, int b, int l, int r, int k) {
        if (r - l == 1) return l;
        int x = nd[nd[b].lc].info - nd[nd[a].lc].info;
        int m = (l + r) / 2;
        if (x >= k) return work(nd[a].lc, nd[b].lc, l, m, k);
        else return work(nd[a].rc, nd[b].rc, m, r, k - x);
    }
    int work(int l, int r, int k) { // [l, r), k > 0
        return s[work(rt[l], rt[r], 0, n, k)];
    }
}; // Dynamic -> Fenwick 包動態開點值域線段樹

```

3.6 Link Cut Tree [d41d8c]

```

template<class Info, class Tag = void>
struct LinkCutTree { // 1-based
    vector<Info> info;
    // vector<Tag> tag;
    vector<array<int, 2>> ch;
    vector<int> p, rev;
    LinkCutTree(int n) : info(n + 1), ch(n + 1), p(n + 1), rev(n + 1) {
        // tag.resize(n + 1);
    }
    bool isrt(int x) {
        return ch[p[x]][0] != x && ch[p[x]][1] != x;
    }
};

```

```

int pos(int x) { // x 是其 par 的左/右
    return ch[p[x]][1] == x;
}
void applyRev(int x) {
    if (!x) return;
    swap(ch[x][0], ch[x][1]);
    rev[x] ^= 1;
}
/*
void apply(int x, const Tag &t) {
    if (!x) return;
    info[x].apply(t);
    tag[x].apply(t);
}
*/
void push(int x) {
    if (rev[x]) {
        applyRev(ch[x][0]);
        applyRev(ch[x][1]);
        rev[x] = 0;
    }
    /*
    apply(ch[x][0], tag[x]);
    apply(ch[x][1], tag[x]);
    tag[x] = Tag();
    */
}
void pull(int x) {
    info[x].pull(info[ch[x][0]], info[ch[x][1]]);
}
void pushAll(int x) {
    if (!isrt(x)) pushAll(p[x]);
    push(x);
}
void rotate
    (int x) { // x 與其 par 交換位置, 只 pull 原本的 par
    int f = p[x], r = pos(x);
    ch[f][r] = ch[x][!r];
    if (ch[x][!r]) p[ch[x][!r]] = f;
    p[x] = p[f];
    if (!isrt(f)) ch[p[f]][pos(f)] = x;
    ch[x][!r] = f, p[f] = x;
    pull(f);
}
void splay(int x) { // x 旋轉到當前的根
    pushAll(x);
    for (int f = p[x]; f = p[x], !isrt(x); rotate(x))
        if (!isrt(f)) rotate(pos(x) == pos(f) ? f : x);
    pull(x);
}
// access(x), access(y) 可以回傳 LCA
int access(int x) { // 根到 x 換成實鏈
    int c;
    for (c = 0; x; c = x, x = p[x]) {
        splay(x);
        ch[x][1] = c;
        pull(x);
    }
    return c;
}
void makeRoot(int x) { // x 變成所在樹的根
    access(x), splay(x), applyRev(x);
}
int findRoot(int x) {
    access(x), splay(x);
    while (ch[x][0]) push(x), x = ch[x][0];
    splay(x); return x;
}
void split(int rt, int x) { // 以 rt 為根, x 為子樹根
    makeRoot(x), access(rt), splay(rt);
}
void link(int rt, int x) { // 以 rt 為根, x 為子樹根
    makeRoot(rt), p[rt] = x;
}
void cut(int rt, int x) { // 以 rt 為根, x 為子樹根
    split(rt, x);
    ch[rt][0] = p[x] = 0;
    pull(rt);
}
bool connected(int x, int y) {
    makeRoot(x);
    return findRoot(y) == x;
}
bool neighbor(int x, int y) {
    return connected(x, y) && p[y] == x && !ch[y][0];
}
void modify(int x, const Info &i) {
    access(x), splay(x);
    info[x] = i, pull(x);
}
/*
void pathApply(int x, int y, const Tag &t) { // x, y 要連通
    split(x, y), apply(x, t);
}
*/
Info pathQuery(int x, int y) { // x, y 要連通
    split(x, y);
    return info[x];
}
}
// Tag: 0b3c49
/* const int Mod = 51061;
struct Tag {
    ll add = 0, mul = 1;
    void apply(const Tag &t) {
        mul = mul * t.mul % Mod;
        add = (add * t.mul % Mod + t.add) % Mod;
    }
}
*/

```

```

}
};
struct Info { /*
    int siz = 0; // 實鏈的長度
    ll val = 0, sum = 0;
    void apply(const Tag &t) {
        val = (val * t.mul % Mod + t.add) % Mod;
        sum = (sum * t.mul % Mod + t.add * siz % Mod) % Mod;
    }
    /*
    void pull(const Info &l, const Info &r) { /*
        siz = 1 + l.siz + r.siz;
        sum = (l.sum + r.sum + val) % Mod;
    }
}
};

```

3.7 LCT Extension [d41d8c]

```

struct Info {
    int siz = 0, vsiz
        = 0, psiz = 0; // 真實子樹大小, 虛子樹大小, 實鏈長度
    ll val = 0, sum = 0, vsum = 0, psum = 0;
    void apply(const Tag &t) {
        val = (val * t.mul % Mod + t.add) % Mod;
        sum = (sum - psum + Mod) % Mod;
        psum = (psum * t.mul % Mod + t.add * psiz % Mod) % Mod;
        sum = (sum + psum) % Mod;
    }
    void pull(const Info &l, const Info &r) {
        psiz = 1 + l.psiz + r.psiz;
        siz = 1 + l.siz + r.siz + vsiz;
        psum = (l.psum + r.psum + val) % Mod;
        sum = (l.sum + r.sum + val + vsum) % Mod;
    }
    void modify(const Info &i) { // 才不會蓋掉 vsum, vsiz
        val = i.val;
    }
    // 下面的 modify 也要改, 不能直接 =
    void operator+=(const Info &i) { // 將實邊轉虛邊時加上資訊
        vsiz += i.siz;
        vsum = (vsum + i.sum) % Mod;
    }
    void operator-=(const Info &i) { // 將虛邊轉實邊時扣除資訊
        vsiz -= i.siz;
        vsum = (vsum - i.sum % Mod + Mod) % Mod;
    }
    Info asSubtree
        () const { // 當前節點作為 Virtual Subtree 根時的狀態
        Info res;
        res.siz = vsiz + 1;
        res.sum = (vsum + val) % Mod;
        return res;
    }
};
struct LinkCutTree { // 加在註解後面
    int access(int x) { // ...splay(x);
        if (ch[x][1]) info[x] += info[ch[x][1]]; // 原右子樹轉虛邊
        if (c) info[x] -= info[c]; // 虛邊轉新右子樹
    }
    void link(int rt, int x) { // 整個換掉
        makeRoot(rt), access(x), splay(x);
        p[rt] = x, info[x] += info[rt]; // rt 成為 x 的虛子樹
        pull(x);
    }
    Info subtreeQuery
        (int rt, int x) { // 以 rt 當 root, x 的 subtree
        assert(connected(rt, x));
        makeRoot(rt);
        access(x), splay(x);
        return info[x].asSubtree();
    }
    Info componentQuery(int x) {
        makeRoot(x);
        return info[x];
    }
};

```

3.8 Treap [d41d8c]

```

template<class Info, class Tag = void>
struct Treap { // 0 -> initial root
    vector<Info> info;
    // vector<Tag> tag;
    vector<int> siz, par, rev, pri;
    vector<array<int, 2>> ch;
    Treap(int n) : info(n + 1), siz
        (n + 1), par(n + 1), rev(n + 1), pri(n + 1), ch(n + 1) {
        // tag.resize(n + 1);
        for (int i = 1; i <= n; i++) siz[i] = 1, pri[i] = rng();
    }
    /*
    void apply(int p, const Tag &t) {
        info[p].apply(siz[p], t);
        tag[p].apply(t);
    }
    */
    void push(int p) {
        if (rev[p]) {
            swap(ch[p][0], ch[p][1]);
            if (ch[p][0]) rev[ch[p][0]] ^= 1;
            if (ch[p][1]) rev[ch[p][1]] ^= 1;
            rev[p] = 0;
        }
    }
}

```

```

    apply(ch[p][0], tag[p]);
    apply(ch[p][1], tag[p]);
    tag[p] = Tag(); */
}
void pull(int p) {
    siz[p] = 1 + siz[ch[p][0]] + siz[ch[p][1]];
    par[ch[p][0]] = par[ch[p][1]] = p;
    info[p].pull(info[ch[p][0]], info[ch[p][1]]);
}
int merge(int a, int b) {
    if (!a || !b) return a ? a : b;
    push(a), push(b);
    if (pri[a] > pri[b]) {
        ch[a][1] = merge(ch[a][1], b);
        pull(a); return a;
    } else {
        ch[b][0] = merge(a, ch[b][0]);
        pull(b); return b;
    }
}
pair<int, int> split(int p, int k) {
    if (!p) return {0, 0};
    push(p);
    if (siz[ch[p][0]] >= k) {
        auto [a, b] = split(ch[p][0], k);
        ch[p][0] = b, pull(p);
        return {a, p};
    } else {
        auto [a, b] = split(ch[p][1], k - siz[ch[p][0]] - 1);
        ch[p][1] = a, pull(p);
        return {p, b};
    }
}
void getArray(int p, vector<Info> &a) {
    if (!p) return;
    push(p);
    getArray(ch[p][0], a);
    a.push_back(info[p]);
    getArray(ch[p][1], a);
}
/*
int getPos(int rt, int p) { // get p's index in array
    int k = siz[ch[p][0]] + 1;
    for (; ; p = par[p]) {
        if (rev[p]) k = siz[p] + 1 - k;
        if (p == rt) return k;
        if (ch[par[p]][1] == p) k += siz[ch[par[p]][0]] + 1;
    }
} // 0a4356
template<class F> int findFirst(int p, F &&pred) {
    if (!p) return 0;
    push(p);
    if (!pred(info[p])) return 0;
    int idx = findFirst(ch[p][0], pred);
    if (!idx)
        idx = 1 + siz[ch[p][0]] + findFirst(ch[p][1], pred);
    return idx;
} */ // 431a65
}; /* // Tag: 7a5a33
struct Tag {
    int setv; ll add;
    void apply(const Tag &t) {
        if (t.setv) {
            setv = t.setv;
            add = t.add;
        } else {
            add += t.add;
        }
    }
}; */
struct Info { /*
    ll val, sum;
    void apply(int siz, const Tag &t) {
        if (t.setv) {
            val = t.setv;
            sum = 1LL * siz * t.setv;
        }
        val += t.add;
        sum += 1LL * siz * t.add;
    } */
    void pull(const Info &l, const Info &r) {
        // sum = val + l.sum + r.sum;
    }
};

```

3.9 KDTree [d41d8c]

```

struct Info {
    static constexpr int DIM = 2;
    array<int, DIM> x, L, R;
    int v = 0, sum = 0;
    void pull(const Info &l, const Info &r) {
        sum = v + l.sum + r.sum;
    }
};
struct KDTree {
    static constexpr int DIM = Info::DIM;
    vector<Info> info;
    vector<int> rt, l, r, p;
    int n, lg;
    KDTree(int n) : n(n),
        lg(__lg(n)), info(1), rt(lg + 1), l(n + 1), r(n + 1) {}
    void pull(int p) {

```

```

        info[p].L = info[p].R = info[p].x;
        info[p].pull(info[l[p]], info[r[p]]);
        for (int ch : {l[p], r[p]}) {
            if (!ch) continue;
            for (int k = 0; k < DIM; k++) {
                info[p].L[k] = min(info[p].L[k], info[ch].L[k]);
                info[p].R[k] = max(info[p].R[k], info[ch].R[k]);
            }
        }
    }
    int rebuild(int l, int r, int dep = 0) {
        if (r == l) return 0;
        int m = (l + r) / 2;
        nth_element(p.begin() + l, p.begin() + m, p.begin() + r,
            [&](int x,
                int y) { return info[x].x[dep] < info[y].x[dep]; });
        int x = p[m];
        this->l[x] = rebuild(l, m, (dep + 1) % DIM);
        this->r[x] = rebuild(m + 1, r, (dep + 1) % DIM);
        pull(x);
        return x;
    }
    void append(int &x) {
        if (!x) return;
        p.push_back(x);
        append(l[x]);
        append(r[x]);
        x = 0;
    }
    void addNode(const Info &i) {
        p.assign(1, info.size());
        info.push_back(i);
        for (int j = 0; j++) {
            if (!rt[j]) {
                rt[j] = rebuild(0, p.size());
                break;
            } else {
                append(rt[j]);
            }
        }
    }
    Info query(int
        p, const array<int, DIM> &l, const array<int, DIM> &r) {
        if (!p) return Info();
        bool inside = true;
        for (int k = 0; k < DIM; k++) {
            inside &= (l[k] <= info[p].L[k] && info[p].R[k] <= r[k]);
        }
        if (inside) return info[p];
        for (int k = 0; k < DIM; k++) {
            if (info[p].R[k] < l[k] || r[k] < info[p].L[k]) {
                return Info();
            }
        }
        Info ans;
        inside = true;
        for (int k = 0; k < DIM; k++) {
            inside &= l[k] <= info[p].x[k] && info[p].x[k] <= r[k];
        }
        if (inside) ans = info[p];
        ans.pull(query(this->l[p], l, r), query(this->r[p], l, r));
        return ans;
    }
    Info query
        (const array<int, DIM> &l, const array<int, DIM> &r) {
        Info res;
        for (int i = 0; i <= lg; i++) {
            res.pull(res, query(rt[i], l, r));
        }
        return res;
    }
};

```

3.10 Sparse Table [d41d8c]

```

template<class T, class F = less<T>> struct RMQ { // [l, r)
    int n;
    F cmp = F();
    vector<vector<T>> g;
    RMQ() {}
    RMQ(const vector<T> &a, F cmp = F()) : cmp(cmp) {
        init(a);
    }
    void init(const vector<T> &a) {
        n = a.size();
        if (!n) return;
        int lg = __lg(n);
        g.resize(lg + 1);
        g[0] = a;
        for (int j = 1; j <= lg; j++) {
            g[j].resize(n - (1 << j) + 1);
            for (int i = 0; i <= n - (1 << j); i++)
                g[j][i] = min
                    (g[j - 1][i], g[j - 1][i + (1 << (j - 1))], cmp);
        }
    }
    T operator()(int l, int r) {
        assert(0 <= l && l < r && r <= n);
        int lg = __lg(r - l);
        return min(g[lg][l], g[lg][r - (1 << lg)], cmp);
    }
};

```

3.11 Mo [d41d8c]

```
struct Query { int id, l, r; };
void mo(vector<Query> &q) {
    int blk = sqrt(q.size());
    sort(q
        .begin(), q.end(), [&](const Query &a, const Query &b) {
            int x = a.l / blk, y = b.l / blk;
            return x == y ? a.r < b.r : x < y;
        });
    int nl = 0, nr = -1;
    for (auto [id, l, r] : qry) {
        while (nr < r) nr++, addR();
        while (l < nl) nl--, addL();
        while (r < nr) delR(), nr--;
        while (nl < l) delL(), nl++;
    }
}
```

3.12 Rollback DSU [d41d8c]

```
struct RollbackDSU {
    int n;
    vector<int> f, siz, stk;
    RollbackDSU(int n) : n(n), f(n), siz(n, 1) {
        tota(f.begin(), f.end(), 0);
    }
    int find(int x) { return x == f[x] ? x : find(f[x]); }
    bool same(int x, int y) { return find(x) == find(y); }
    bool merge(int x, int y) {
        x = find(x), y = find(y);
        if (x == y) return false;
        if (siz[x] < siz[y]) swap(x, y);
        siz[x] += siz[y];
        f[y] = x;
        n--;
        stk.push_back(y);
        return true;
    }
    void undo(int x) {
        while (stk.size() > x) {
            int y = stk.back();
            stk.pop_back();
            n++;
            siz[f[y]] -= siz[y];
            f[y] = y;
        }
    }
    int size(int x) { return siz[find(x)]; }
};
```

4 Flow Matching

4.1 Dinic [d41d8c]

```
template<class T> struct Dinic {
    // augment time: O(VE), O(E) for unit capacity,
    // augment number: min(O(F), O(V))
    // min(O(E^0.5), O(V^2/3)) for unit capacity
    // so O(V^0.5) for deg_in(u) or deg_out(u) <= 1
    // so bipartite matching: O(EV^0.5)
    struct Edge {
        int to;
        T f, cap; // 流量跟容量
    };
    int n, m, s, t;
    const T INF_FLOW = numeric_limits<T>::max() / 2;
    vector<vector<int>> g;
    vector<Edge> e;
    vector<int> h, cur;
    Dinic(int n) : n(n), m(0), g(n), h(n), cur(n) {}
    void addEdge(int u, int v, T cap) {
        e.push_back({v, 0, cap});
        e.push_back({u, 0, 0});
        g[u].push_back(m++);
        g[v].push_back(m++);
    }
    bool bfs() {
        fill(h.begin(), h.end(), -1);
        h[s] = 0; queue<int> q;
        q.push(s);
        while (!q.empty()) {
            int u = q.front(); q.pop();
            for (int id : g[u]) {
                auto [v, f, cap] = e[id];
                if (f != cap && h[v] == -1) {
                    h[v] = h[u] + 1;
                    if (v == t) return true;
                    q.push(v);
                }
            }
        }
        return false;
    }
    T dfs(int u, T flow) {
        if (flow == 0) return 0;
        if (u == t) return flow;
        for (int &i = cur[u]; i < g[u].size(); i++) {
            int j = g[u][i];
            auto [v, f, cap] = e[j];
            if (f == cap || h[u] + 1 != h[v]) continue;
            T mn = dfs(v, min(flow, cap - f));
            f += mn;
            if (mn > 0) {
                e[j].f += mn;
                e[j ^ 1].f -= mn;
                return mn;
            }
        }
        return 0;
    }
    T work(int s_, int t_) {
        s = s_; t = t_; T f = 0;
        while (bfs()) {
            fill(cur.begin(), cur.end(), 0);
            while (true) {
                T res = dfs(s, INF_FLOW);
                if (res == 0) break;
                f += res;
            }
            return f;
        }
    }
    /*
    void reuse(int n_) { // 走殘留網路, res += f
        for (; n < n_; n++) {
            g.emplace_back();
            h.emplace_back();
            cur.emplace_back();
        }
    } */
};
```

```
if (mn > 0) {
    e[j].f += mn;
    e[j ^ 1].f -= mn;
    return mn;
}
}
return T{};
}
T work(int s_, int t_) {
    s = s_; t = t_; T f = 0;
    while (bfs()) {
        fill(cur.begin(), cur.end(), 0);
        while (true) {
            T res = dfs(s, INF_FLOW);
            if (res == 0) break;
            f += res;
        }
        return f;
    }
}
/*
void reuse(int n_) { // 走殘留網路, res += f
    for (; n < n_; n++) {
        g.emplace_back();
        h.emplace_back();
        cur.emplace_back();
    }
} */
};
```

4.2 Min Cut [d41d8c]

```
void policeChase(int n, Dinic<int> &d) {
    int ans = d.work(0, n - 1);
    vector<int> vis(n);
    [&](this auto &&self, int u) -> void {
        if (vis[u]) return;
        vis[u] = 1;
        for (int id : d.g[u]) {
            auto [to, f, cap] = d.e[id];
            if (cap - f > 0) self(to);
        }
    } (0);
    for (int i = 0; i < n; i++) {
        if (!vis[i]) continue;
        for (int id : d.g[i]) {
            if (id & 1) continue;
            auto e = d.e[id];
            if (!vis[e.to]) cout << i + 1 << " " << e.to + 1 << "\n";
        }
    }
}
```

4.3 MCMF [d41d8c]

```
template<class Tf, class Tc> struct MCMF {
    struct Edge {
        int to;
        Tf f, cap; // 流量跟容量
        Tc cost;
    };
    int n, m, s, t;
    const Tf INF_FLOW = numeric_limits<Tf>::max() / 2;
    const Tc INF_COST = numeric_limits<Tc>::max() / 2;
    vector<Edge> e;
    vector<vector<int>> g;
    vector<Tc> dis;
    vector<int> rt, inq;
    MCMF(int n) : n(n), m(0), g(n) {}
    void addEdge(int u, int v, Tf cap, Tc cost) {
        e.push_back({v, 0, cap, cost});
        e.push_back({u, 0, 0, -cost});
        g[u].push_back(m++);
        g[v].push_back(m++);
    }
    bool spfa() { // O(VE)
        dis.assign(n, INF_COST);
        rt.assign(n, -1), inq.assign(n, 0);
        queue<int> q; q.push(s);
        dis[s] = 0, inq[s] = 1;
        while (!q.empty()) {
            int u = q.front(); q.pop();
            inq[u] = 0;
            for (int id : g[u]) {
                auto [v, f, cap, cost] = e[id];
                Tc ndis = dis[u] + cost;
                if (f < cap && dis[v] > ndis) {
                    dis[v] = ndis, rt[v] = id;
                    if (!inq[v])
                        q.push(v), inq[v] = 1;
                }
            }
        }
        return dis[t] != INF_COST;
    }
    // spfa: O(FVE), dijkstra: O(VE + FE log V)
    pair<Tf, Tc> work(int s_, int t_, Tf need) {
        s = s_, t = t_;
        Tf flow{}; Tc cost{};
        while (
            spfa() { // dijkstra: flow == 0 ? spfa() : dijkstra()
                Tf f = need;
                for (int i = t; i != s; i = e[rt[i] ^ 1].to)
                    f = min(f, e[i].cap - e[i].f);
                if (f == 0) break;
                for (int i = t; i != s; i = e[rt[i] ^ 1].to)
                    e[i].f += f, e[i ^ 1].f -= f;
                flow += f, cost += f * e[rt[t]].cost;
            }
        );
        return {flow, cost};
    }
};
```

```

    f = min(f, e[rt[i]].cap - e[rt[i]].f);
    for (int i = t; i != s; i = e[rt[i]] ^ 1).to)
        e[rt[i]].f += f, e[rt[i]] ^ 1.f -= f;
    flow += f, need -= f;
    cost += f * dis[t];
    if (need == 0) break;
}
return {flow, cost};
} /*
vector<Tc> pot;
bool dijkstra
() { // 0(ElogV), 用上一輪的 dis 當 pot, 邊權變非負
    pot = dis;
    dis.assign(n, INF_COST), rt.assign(n, -1);
    priority_queue
        <pair<Tc, int>, vector<pair<Tc, int>>, greater<>> pq;
    dis[s] = 0;
    pq.emplace(dis[s], s);
    while (!pq.empty()) {
        auto [d, u] = pq.top(); pq.pop();
        if (dis[u] < d) continue;
        for (int id : g[u]) {
            auto [v, f, cap, cost] = e[id];
            Tc ndis = dis[u] + cost + pot[u] - pot[v];
            if (f < cap && dis[v] > ndis) {
                dis[v] = ndis, rt[v] = id;
                pq.emplace(ndis, v);
            }
        }
    }
    for (int i = 0; i < n; i++)
        if (dis[i] != INF_COST) dis[i] += pot[i];
    return dis[t] != INF_COST;
} */ // 54295b
}; // a224e5

```

4.4 Hungarian [d41d8c]

```

struct Hungarian { // 0-based, 0(VE)
    int n, m;
    vector<vector<int>> adj;
    vector<int> used, vis;
    vector<pair<int, int>> match;
    Hungarian(int n, int m) : n(n), m(m) {
        adj.assign(n + m, {});
        used.assign(n + m, -1);
        vis.assign(n + m, 0);
    }
    void addEdge(int u, int v) {
        adj[u].push_back(n + v);
        adj[n + v].push_back(u);
    }
    bool dfs(int u) {
        int sz = adj[u].size();
        for (int i = 0; i < sz; i++) {
            int v = adj[u][i];
            if (vis[v] == 0) {
                vis[v] = 1;
                if (used[v] == -1 || dfs(used[v])) {
                    used[v] = u;
                    return true;
                }
            }
        }
        return false;
    }
    vector<pair<int, int>> work() {
        match.clear();
        used.assign(n + m, -1);
        vis.assign(n + m, 0);
        for (int i = 0; i < n; i++) {
            fill(vis.begin(), vis.end(), 0);
            dfs(i);
        }
        for (int i = n; i < n + m; i++)
            if (used[i] != -1)
                match.emplace_back(used[i], i - n);
        return match;
    }
};

```

• 有向無環圖:

- 最小不相交路徑覆蓋: 最小路徑數 = 頂點數 - 最大匹配數
- 最小相交路徑覆蓋: 先用 Floyd 求傳遞封包, 有連邊就建邊, 然後再套最小不相交路徑覆蓋

• 二分圖:

- 最小點覆蓋: 選出一些點, 使每條邊至少有一個端點被選到的最少數量
- 最小點覆蓋 = 最大匹配數
- 還原解: flow 的作法是從源點開始 dfs, 只走 $cap - flow > 0$ 的邊, 最後挑選左邊還沒被跑過的點和右邊被跑過的點當作覆蓋的點
- 最大獨立集: 選出一些點, 使這些點兩兩沒有邊連接的最大數量
- 最大獨立集 = 點數 - 最大匹配數
- 還原解: 最小點覆蓋的補集
- 最少邊覆蓋: 選出一些邊, 讓所有點都被覆蓋到的最少數量
- 最少邊覆蓋 = 點數 - 孤立點數 - 最大匹配數
- 還原解: 最大匹配 + 未被匹配的非孤立點隨便配

5 String

5.1 Hash [234076]

```

const int D = 59;
vector<int> rollingHash(string &s) {
    vector<int> a{0};
    for (auto c : s)
        a.push_back(mul(a.back(), D) + (c - 'A' + 1));
    return a;
}
int qryHash(vector<int> &h, int l, int r) { // [l, r)
    return sub(h[r], mul(h[l], power(D, r - l)));
}

```

5.2 KMP [cdf28b]

```

template<class T> struct KMP {
    int n;
    T pat;
    vector<int> f;
    // f 存匹配失敗時, 移去哪
    // 也就是 pat[0, i] 的最長共同前後綴長度
    // ex : a b c a b c
    //      -1 -1 -1 0 1 2
    KMP(const T &pat) : n(pat.size()), pat(pat), f(n, -1) {
        for (int i = 1; i < n; i++) {
            int j = f[i - 1];
            while (j != -1 && pat[j + 1] != pat[i]) j = f[j];
            if (pat[j + 1] == pat[i]) f[i] = j + 1;
        }
    }
    vector<int> work(const T &s) {
        vector<int> pos;
        for (int i = 0, j = -1; i < s.size(); i++) {
            while (j != -1 && pat[j + 1] != s[i]) j = f[j];
            if (pat[j + 1] == s[i]) j++;
            if (j + 1 == n) {
                pos.push_back(i - j);
                j = f[j];
            }
        }
        return pos;
    }
};

```

5.3 Z Function [a1f65d]

```

// z[i] 表示 s 和 s[i, n - 1] (以 s[i] 開頭的后綴)
// 的最長公共前綴 (LCP) 的長度
template<class T> vector<int> Z(const T &s) {
    int n = s.size();
    vector<int> z(n);
    z[0] = n; // lcp(s, s), -1 or n
    for (int i = 1, j = 1; i < n; i++) {
        z[i] = max(0, min(j + z[j] - i, z[i - j]));
        while (i + z[i] < n && s[z[i]] == s[i + z[i]]) z[i]++;
        if (i + z[i] > j + z[j]) j = i;
    }
    return z;
}

```

5.4 Manacher [9f76ee]

```

// 找到對於每個位置的迴文半徑
template<class T> vector<int> manacher(const T &s) {
    vector<int> t{-1};
    for (auto c : s) t.push_back(c), t.push_back(-1);
    int n = t.size();
    vector<int> r(n);
    for (int i = 0,
        j = 0; i < n; i++) { // i 是中心, j 是最長回文字串中心
        if (2 * j - i >= 0 && j + r[j] > i)
            r[i] = min(r[2 * j - i], j + r[j] - i);
        while (i - r[i]
            >= 0 && i + r[i] < n && t[i - r[i]] == t[i + r[i]])
            r[i]++;
        if (i + r[i] > j + r[j]) j = i;
    }
    return r;
}
// # a # b # a #
// 1 2 1 4 1 2 1
// # a # b # b # a #
// 1 2 1 2 5 2 1 2 1
// 值 -1 代表原回文字串長度
// (id - val + 1) / 2 可得原字串回文開頭

```

5.5 Trie [595441]

```

const int N = 1E7; // 0 -> initial state
const int ALPHABET_SIZE = 26;
int tot = 0;
int trie[N][ALPHABET_SIZE], cnt[N];
int newNode() {
    tot++;
    fill_n(trie[tot], ALPHABET_SIZE, 0);
    cnt[tot] = 0;
    return tot;
}

```

```

void reset() {
    tot = -1;
    newNode();
}
void add(const string &s, int i = 0, int p = 0) {
    if (i == s.size()) {
        cnt[p]++;
        return;
    }
    int &q = trie[p][s[i] - 'a'];
    if (!q) q = newNode();
    add(s, i + 1, q);
}
int find(const string &s, int i = 0, int p = 0) {
    if (i == s.size()) return cnt[p];
    int q = trie[p][s[i] - 'a'];
    if (!q) return 0;
    return find(s, i + 1, q);
}

```

5.6 SA_[9262af]

```

template<class T> struct SuffixArray {
    int n;
    vector<int> sa, rk, lc;
    // n: 字串長度
    // sa: 後綴數組, sa[i] 表示第 i 小的後綴的起始位置
    // rk: 排名數組, rk[i] 表示從位置 i 開始的後綴的排名
    // lc: LCP
    // 數組, lc[i] 表示 sa[i] 和 sa[i + 1] 的最長公共前綴長度
    SuffixArray(
        const T &s : n(s.size()), sa(n), lc(n - 1), rk(n) {
        iota(sa.begin(), sa.end(), 0);
        sort(sa.begin(), sa.end(), [&](int a, int b) { return s[a] < s[b]; });
        rk[sa[0]] = 0;
        for (int i = 1; i < n; i++)
            rk[sa[i]] = rk[sa[i - 1]] + (s[sa[i]] != s[sa[i - 1]]);
        int k = 1;
        vector<int> tmp, cnt(n);
        tmp.reserve(n);
        while (rk[sa[n - 1]] < n - 1) {
            tmp.clear();
            for (int i = 0; i < k; i++) tmp.push_back(n - k + i);
            for (auto i : sa) if (i >= k) tmp.push_back(i - k);
            fill(cnt.begin(), cnt.end(), 0);
            for (int i = 0; i < n; i++) cnt[rk[i]]++;
            for (int i = 1; i < n; i++) cnt[i] += cnt[i - 1];
            for (int i =
                n - 1; i >= 0; i--) sa[--cnt[rk[tmp[i]]]] = tmp[i];
            swap(rk, tmp); rk[sa[0]] = 0;
            for (int i = 1; i < n; i++) rk[sa[i]] = rk[
                sa[i - 1]] + (tmp[sa[i - 1]] < tmp[sa[i]] || sa[i -
                1] + k == n || tmp[sa[i - 1] + k] < tmp[sa[i] + k]);
            k *= 2;
        }
        for (int i = 0, j = 0; i < n; i++) {
            if (rk[i] == 0) {
                j = 0;
            } else {
                for (j -= j > 0; i + j < n && sa[rk[i] - 1]
                    + j < n && s[i + j] == s[sa[rk[i] - 1] + j]; j++);
                lc[rk[i] - 1] = j;
            }
        }
    }; // ad4f20
    RMQ<int> rmq(sa.lc);
    auto lcp = [&](int i, int j) { // [i, j]
        i = sa.rk[i], j = sa.rk[j];
        if (i > j) swap(i, j);
        assert(i != j);
        return rmq(i, j);
    };
};

```

5.7 AC_[d34486]

```

struct AC {
    static constexpr int ALPHABET_SIZE = 26;
    struct Node {
        int fail; // 指向最長後綴
        int cnt; // 有多少模式字串是自己的後綴
        array<int, ALPHABET_SIZE> ch, next;
        // next 是補全後的轉移
    };
    vector<Node> t;
    AC() : t(1) {}
    int newNode() {
        t.emplace_back();
        return t.size() - 1;
    }
    int insert(const string &s) {
        int u = 0;
        for (char c : s) {
            if (!t[u].ch[c - 'a']) {
                int v = newNode();
                t[u].ch[c - 'a'] = v;
            }
            u = t[u].ch[c - 'a'];
        }
        t[u].cnt++;
    }
};

```

```

        return u;
    }
    void build() {
        queue<int> q;
        for (int c = 0; c < ALPHABET_SIZE; c++) {
            if (t[0].ch[c]) {
                q.push(t[0].ch[c]);
                t[0].next[c] = t[0].ch[c];
            }
        }
        while (!q.empty()) {
            int u = q.front(); q.pop();
            for (int c = 0; c < ALPHABET_SIZE; c++) {
                if (t[u].ch[c]) {
                    int v = t[u].ch[c];
                    t[v].fail = t[t[u].fail].next[c];
                    t[v].cnt += t[t[u].fail].cnt;
                    t[u].next[c] = v;
                    q.push(v);
                } else {
                    t[u].next[c] = t[t[u].fail].next[c];
                }
            }
        }
    }
};

```

5.8 SAM_[83a519]

```

struct SAM {
    // 0 -> initial state
    static constexpr int ALPHABET_SIZE = 26;
    // node -> strings with the same endpos set
    // link -> longest suffix with different endpos set
    // len -> state's longest suffix
    // fpos -> first endpos
    // strlen range -> [len(link) + 1, len]
    struct Node {
        int len, link = -1, fpos;
        array<int, ALPHABET_SIZE> next;
    };
    vector<Node> t;
    SAM() : t(1) {}
    int newNode() {
        t.emplace_back();
        return t.size() - 1;
    }
    int extend(int p, int c) {
        int cur = newNode();
        t[cur].len = t[p].len + 1;
        t[cur].fpos = t[cur].len - 1;
        while (p != -1 && !t[p].next[c]) {
            t[p].next[c] = cur;
            p = t[p].link;
        }
        if (p == -1) {
            t[cur].link = 0;
        } else {
            int q = t[p].next[c];
            if (t[p].len + 1 == t[q].len) {
                t[cur].link = q;
            } else {
                int r = newNode();
                t[r] = t[q];
                t[r].len = t[p].len + 1;
                while (p != -1 && t[p].next[c] == q) {
                    t[p].next[c] = r;
                    p = t[p].link;
                }
                t[q].link = t[cur].link = r;
            }
        }
        return cur;
    }
}; // 57198e
void solve(int n, string s, ll k) { // Substring Order II
    vector<int> last(n + 1);
    SAM sam;
    for (int i = 0; i < n; i++)
        last[i + 1] = sam.extend(last[i], s[i] - 'a');
    int sz = sam.t.size();

    vector<int> cnt(sz); // endpos size
    for (int i = 1; i <= n; i++) cnt[last[i]]++;
    vector<vector<int>> g(sz);
    for (int i = 1; i < sz; i++)
        g[sam.t[i].link].push_back(i);
    [&](this auto &&self, int u) -> void {
        for (auto v : g[u]) self(v), cnt[u] += cnt[v];
    }(0);

    vector<ll> dp(sz, -1);
    // for any path from
    // root, how many substring's prefix is the the path string
    [&](this auto &&self, int u) -> ll {
        if (dp[u] != -1) return dp[u];
        dp[u] = cnt[u]; // distinct = 1
        for (int c = 0; c < SAM::ALPHABET_SIZE; c++) {
            int v = sam.t[u].next[c];
            if (v) dp[u] += self(v);
        }
        return dp[u];
    };
};

```

```

} (0);

int p = 0; string ans;
while (k > 0) { // 1-based
    for (int c = 0; c < SAM::ALPHABET_SIZE; c++) {
        int v = sam.t[p].next[c];
        if (v) {
            if (k > dp[v]) {
                k -= dp[v];
            } else {
                ans.push_back('a' + c);
                k -= cnt[v]; // distinct: --
                p = v; break;
            }
        }
    }
}
cout << ans << "\n";
}

```

5.9 Palindrome Tree [1da7bb]

```

struct PAM {
    // 0 -> even root, 1 -> odd root
    static constexpr int ALPHABET_SIZE = 26;
    // fail -> longest prefix(suffix) palindrome
    // number end at i = end at link[last[i]] + 1
    struct Node {
        int len, fail, cnt;
        array<int, ALPHABET_SIZE> next;
        Node() : len{0}, fail{0}, next{} {}
    };
    vector<int> s;
    vector<Node> t;
    PAM() {
        t.assign(2, Node());
        t[0].len = 0, t[0].fail = 1;
        t[1].len = -1;
    }
    int newNode() {
        t.emplace_back();
        return t.size() - 1;
    }
    int getFail(int p, int i) {
        while (i - t[p].len < 1 || s[i - t[p].len - 1] != s[i])
            p = t[p].fail;
        return p;
    }
    int extend(int p, int c) {
        int i = s.size();
        s.push_back(c);
        p = getFail(p, i);
        if (!t[p].next[c]) {
            int r = newNode();
            int v = getFail(t[p].fail, i);
            t[r].len = t[p].len + 2;
            t[r].fail = t[v].next[c];
            t[p].next[c] = r;
        }
        return p = t[p].next[c];
    }
};

// string s; cin >> s;
// int n = s.length();
// vector<int> last(n + 1);
// last[0] = 1;
// PAM pam;
// for (int i = 0; i < n; i++)
//     last[i + 1] = pam.extend(last[i], s[i] - 'a');
// int sz = pam.t.size();
// vector<int> cnt(sz);
// for (int i = 1; i <= n; i++)
//     cnt[last[i]]++; // 去重 = 1
// for (int i = sz - 1; i > 1; i--)
//     cnt[pam.t[i].fail] += cnt[i];

```

5.10 Duval [f0a5c2]

```

// duval_algorithm
// 將字串分解成若干個非嚴格遞減的非嚴格遞增字串
template<class T> vector<T> duval(const T &s) {
    int i = 0, n = s.size();
    vector<T> res;
    while (i < n) {
        int k = i, j = i + 1;
        for (; j < n && s[k] <= s[j]; j++) {
            if (s[k] < s[j]) k = i;
            else k++;
        }
        while (i <= k) {
            res.emplace_back(s.begin() + i, s.begin() + i + j - k);
            i += j - k;
        }
    }
    return res;
}

// 最小旋轉字串
template<class T> T minRound(T s) {
    s.insert(s.end(), s.begin(), s.end());
    int i = 0, n = s.size(), start = i;
    while (i < n / 2) {
        start = i;

```

```

        int k = i, j = i + 1;
        for (; j < n && s[k] <= s[j]; j++) {
            if (s[k] < s[j]) k = i;
            else k++;
        }
        while (i <= k) i += j - k;
    }
    return {s.begin() + start, s.begin() + start + n / 2};
}

```

6 Math

6.1 Mint [78bb96]

```

// P * P > ll => mul in pollard rho
// P * 2 > int => x int to ll
const int P = 1E9 + 7;
template<class T> constexpr T power(T a, ll b) {
    T res{1};
    for (; b > 0; b >>= 1, a *= a)
        if (b & 1) res *= a;
    return res;
} // b2c151

struct Z { // simple
    int x;
    Z(ll v = 0) : x(v % P) { if (x < 0) x += P; }
    Z operator-() const { return P - x; }
    Z inv() const { return power(*this, P - 2); }

    Z &operator
    +=(Z a) { if ((x += a.x) >= P) x -= P; return *this; }
    Z &operator
    -=(Z a) { if ((x -= a.x) < 0) x += P; return *this; }
    Z &operator*=(Z a) { x = 1LL * x * a.x % P; return *this; }
    Z &operator/=(Z a) { return *this *= a.inv(); }

    friend Z operator+(Z a, Z b) { return a += b; }
    friend Z operator-(Z a, Z b) { return a -= b; }
    friend Z operator*(Z a, Z b) { return a *= b; }
    friend Z operator/(Z a, Z b) { return a /= b; }
} /* friend istream &operator>>(istream &is, Z &a) {
    ll v; is >> v;
    a = v; return is;
} // 4a6a72
friend ostream &operator<<(ostream &os, Z a) {
    return os << a.x;
} // 81cfbc */
}; // f00eb5

template<int P> struct Mint {
    static inline int Mod = 998244353;
    static int p() { return P > 0 ? P : Mod; }
    static void setMod(int Mod_) { Mod = Mod_; }
    int x;
    Mint(ll v = 0) : x(v % p()) { if (x < 0) x += p(); }
    Mint operator-() const { return p() - x; }
    Mint inv() const { return power(*this, p() - 2); }

    Mint &operator+=(Mint
        a) { if ((x += a.x) >= p()) x -= p(); return *this; }
    Mint &operator-=(
        Mint a) { if ((x -= a.x) < 0) x += p(); return *this; }
    Mint &operator
    *=(Mint a) { x = 1LL * x * a.x % p(); return *this; }
    Mint &operator/=(Mint a) { return *this *= a.inv(); }

    friend Mint operator+(Mint a, Mint b) { return a += b; }
    friend Mint operator-(Mint a, Mint b) { return a -= b; }
    friend Mint operator*(Mint a, Mint b) { return a *= b; }
    friend Mint operator/(Mint a, Mint b) { return a /= b; }
} /* friend istream &operator>>(istream &is, Mint &a) {
    ll v; is >> v;
    a = v; return is;
} // f964a7
friend ostream &operator<<(ostream &os, Mint a) {
    return os << a.x;
} // d01053 */
};
using Z = Mint<P>; // 426039

vector<Z> fac, invfac, inv;
void init(int n) {
    fac.resize(n + 1);
    invfac.resize(n + 1);
    inv.resize(n + 1);
    fac[0] = invfac[0] = 1;
    for (int i = 1; i <= n; i++) {
        fac[i] = fac[i - 1] * i;
    }
    invfac[n] = fac[n].inv();
    for (int i = n; i > 0; i--) {
        invfac[i - 1] = invfac[i] * i;
        inv[i] = invfac[i] * fac[i - 1];
    }
} // c5dd4f

Z binom(int n, int m) {
    if (n < m || m < 0) return 0;
    return fac[n] * invfac[m] * invfac[n - m];
} // 9326d0

Z lucas(ll n, ll m) { // O(p + T log(n)), p is prime
    return m ? binom(n % P, m % P) * lucas(n / P, m / P) : 1;
} // de3d87

```

6.2 Sieve [ad6c6e]

```
vector<int> minp, primes, mu, phi, dnum, pnum;
vector<ll> dsum;
void sieve(int n) {
    minp.assign(n + 1, 0);
    phi.assign(n + 1, 0), mu.assign(n + 1, 0);
    phi[1] = mu[1] = 1;
    for (int i = 2; i <= n; i++) {
        if (!minp[i]) {
            minp[i] = i;
            primes.push_back(i);
            phi[i] = i - 1;
            mu[i] = -1;
        }
        for (int p : primes) {
            if (i * p > n) break;
            minp[i * p] = p;
            if (p == minp[i]) {
                phi[i * p] = phi[i] * p;
                mu[i * p] = 0;
                break;
            } else {
                phi[i * p] = phi[i] * (p - 1);
                mu[i * p] = -mu[i];
            }
        }
    }
    dnum.assign(n + 1, 0), dsum.assign(n + 1, 0);
    for (int i = 1; i <= n; i++)
        for (int j = i; j <= n; j += i) dnum[j]++, dsum[j] += i;
    pnum.assign(n + 1, 0);
    for (int p : primes) for (int j = p; j <= n; j += p) pnum[j]++;
}
```

6.3 Matrix [106d8e]

```
template<class T> struct Mat {
    int n = 0, m = 0; // n == 0: identity of any size
    vector<vector<T>> a;
    Mat(int) {} // for power's T res[1]
    Mat(int n, int m) : n(n), m(m), a(n, vector<T>(m)) {}
    Mat(const vector<vector<T>> &a) : n(a.size()), m(a[0].size()), a(a) {}
    Mat operator*(const Mat &b) const {
        if (!n) return b;
        if (!b.n) return *this;
        assert(m == b.n);
        Mat res(n, b.m);
        for (int i = 0; i < n; i++)
            for (int l = 0; l < m; l++)
                for (int j = 0; j < b.m; j++)
                    res.a[i][j] += a[i][l] * b.a[l][j];
        return res;
    }
    Mat &operator*=(const Mat &b) { return *this = *this * b; }
};
```

6.4 Miller Rabin Pollard Rho [394cfb]

```
ll mul(ll a, ll b, ll p) {
    ll res = a * b - ll(1.L * a * b / p) * p;
    res %= p;
    if (res < 0) res += p;
    return res;
}
ll power(ll a, ll b, ll p) {
    ll res = 1;
    for (; b; b /= 2, a = mul(a, a, p))
        if (b & 1) res = mul(res, a, p);
    return res;
}
vector<ll> chk{2, 325, 9375, 28178, 450775, 9780504, 1795265022};
bool check(ll a, ll d, int s, ll n) {
    a = power(a, d, n);
    if (a <= 1) return 1;
    for (int i = 0; i < s; i++, a = mul(a, a, n)) {
        if (a == 1) return 0;
        if (a == n - 1) return 1;
    }
    return 0;
}
bool isPrime(ll n) {
    if (n < 2) return 0;
    if (n % 2 == 0) return n == 2;
    ll d = n - 1, s = 0;
    while (d % 2 == 0) d /= 2, s++;
    for (ll i : chk)
        if (!check(i, d, s, n)) return 0;
    return 1;
}
const vector<ll> small = {2, 3, 5, 7, 11, 13, 17, 19};
ll findFactor(ll n) {
    if (isPrime(n)) return 1;
    for (ll p : small)
        if (n % p == 0) return p;
    ll x, y = 2, d, t = 1;
    auto f = [&](ll a) {
        return (mul(a, a, n) + t) % n;
    };
};
```

```
for (int l = 2; ; l *= 2) {
    x = y;
    int m = min(l, 32);
    for (int i = 0; i < l; i += m) {
        d = 1;
        for (int j = 0; j < m; j++)
            y = f(y), d = mul(d, abs(x - y), n);
        ll g = __gcd(d, n);
        if (g == n) {
            l = 1, y = 2, ++t;
            break;
        }
        if (g != 1) return g;
    }
}
map<ll, int> res;
void pollardRho(ll n) {
    if (n == 1) return;
    if (isPrime(n)) {
        res[n]++;
        return;
    }
    ll d = findFactor(n);
    pollardRho(n / d), pollardRho(d);
}
```

6.5 Primitive Root [b0ba96]

```
int findPrimitiveRoot(ll m) {
    MLong<0>::setMod(m); // MLong if needed
    ll phi = m; // m - 1 if m prime
    res.clear();
    pollardRho(m);
    for (auto [p, _] : res) phi = phi / p * (p - 1);
    vector<ll> pr; // prime factors of phi
    res.clear();
    pollardRho(phi);
    for (auto &[p, _] : res) pr.push_back(p);
    for (int i = 1; i <= m; i++) {
        bool ok = true;
        for (int j = 0; j < int(pr.size()) && ok; j++)
            ok &= power(MLong<0>(i), phi / pr[j]).x != 1;
        if (ok) return i;
    }
    return -1; // m != 1, 2, 4, p^k, 2(p^k)
}
```

6.6 CRT [35109b]

```
// ax = b (mod m) 的前提是 gcd(a, m) | b
// Ax + By = C (g = gcd(A, B))
// (A/g) * x + (B/g) * y = C/g
// solve (A/g) * x' + (B/g) * y' = 1
// x = (x' * (C/g)) + i(B/g)
pair<ll, ll> exgcd(ll a, ll b) {
    if (b == 0) return {1, 0};
    auto [y, x] = exgcd(b, a % b);
    return {x, y - (a / b) * x};
} // 結果可能為負，注意正規化
// Smallest non-negative solution
using i128 = __int128_t;
pair<ll, ll> CRT(ll r1, ll m1, ll r2, ll m2) {
    ll g = __gcd(m1, m2);
    if ((r2 - r1) % g) return {-1, g};
    m1 /= g, m2 /= g;
    auto [p1, p2] = exgcd(m1, m2);
    i128 lcm = i128(m1) * m2 * g;
    i128 res = i128(p1) * (r2 - r1) * m1 + r1;
    return {(res % lcm + lcm) % lcm, lcm};
} // 8ddc68
pair<ll, ll> EXCRT(vector<pair<int, int>> a) {
    ll R = 0, M = 1;
    for (auto [r, m] : a) {
        auto [res, lcm] = CRT(R, M, r, m);
        if (res == -1) return {-1, -1};
        R = res, M = lcm;
    }
    return {R, M};
} // 3396d1
// gcd(mod) = 1, support 3 1E9 Mod
pair<i128, i128> CRT(vector<pair<int, int>> a) {
    i128 s = 1, res = 0;
    for (auto [r, m] : a) s *= m;
    for (auto [r, m] : a) {
        i128 t = s / m;
        res = (res + r * t % s * exgcd(t, m).first % s) % s;
    }
    return {(res + s) % s, s};
} // 8ffc2d
```

6.7 XOR Basis [02f0c0]

```
auto add = [&](vector<int> &bas, int x) {
    for (auto i : bas) x = min(x, x ^ i);
    if (x) bas.push_back(x);
};
sort(bas.begin(), bas.end()); // 最簡化列梯
for (auto i = bas.begin(); i != bas.end(); i++) {
    for (auto j = next(i); j != bas.end(); j++) {
        *j = min(*j, *j ^ *i);
    }
}
```

```

}
}
// [l, r] 的區間 xor max (1-indexed)
vector<array<int, B>> bas(n + 1), lst(n + 1);
for (int i = 1; i <= n; i++) {
    int x = a[i - 1], p = i;
    bas[i] = bas[i - 1], lst[i] = lst[i - 1];
    for (int j = B - 1; j >= 0; j--) {
        if (x >> j & 1) {
            if (!bas[i][j]) {
                bas[i][j] = x, lst[i][j] = p;
                break;
            }
            if (lst[i][j] < p)
                swap(lst[i][j], p), swap(x, bas[i][j]);
            x ^= bas[i][j];
        }
    }
}
auto qry = [&](int l, int r) -> int {
    int mx = 0;
    for (int i = B - 1; i >= 0; i--)
        if (lst[r][i] >= l)
            mx = max(mx, mx ^ bas[r][i]);
    return mx;
};

```

6.8 Gaussian Elimination [5d1aa7]

```

// 找反矩陣
就開 2n · 右邊放單位矩陣，做完檢查左半是不是單位，回傳右半
// 0 : no solution
// -1 : infinity solution
// 1 : one solution
template<class T> tuple<T,
    int, vector<T>> gaussianElimination(vector<vector<T>> a) {
    T det = 1;
    bool zeroDet = false;
    int n = a.size(), m = a[0].size(), rk = 0, sgn = 1;
    for (int c = 0; c < n; c++) {
        int p = -1;
        for (int r = rk; r < n; r++) {
            if (a[r][c] != 0) {
                p = r;
                break;
            }
        }
        if (p == -1) {
            zeroDet = true;
            continue;
        }
        if (p != rk) swap(a[rk], a[p]), sgn *= -1;
        det *= a[rk][c];
        T inv = 1 / a[rk][c];
        for (int j = c; j < m; j++) a[rk][j] *= inv;
        for (int r = 0; r < n; r++) {
            if (r == rk || a[r][c] == 0) continue;
            T fac = a[r][c];
            for (int j = c; j < m; j++)
                a[r][j] -= fac * a[rk][j];
        }
        rk++;
    }
    det = (zeroDet ? 0 : det * sgn);
    for (int r = rk; r < n; r++)
        if (a[r][m - 1] != 0) return {det, 0, {}};
    if (rk < n) return {det, -1, {}};
    vector<T> ans(n);
    for (int i = 0; i < n; i++) ans[i] = a[i][m - 1];
    return {det, 1, ans};
} // 0064b9
template<class T> tuple<int, vector
    <T>, vector<vector<T>>> findBasis(vector<vector<T>> a) {
    int n = a.size(), m = a[0].size(), rk = 0;
    vector<int> pos(m - 1, -1);
    for (int c = 0; c < m - 1; c++) {
        int p = -1;
        for (int r = rk; r < n; r++) {
            if (a[r][c] != 0) {
                p = r;
                break;
            }
        }
        if (p == -1) continue;
        if (p != rk) swap(a[rk], a[p]);
        pos[c] = rk;
        T inv = 1 / a[rk][c];
        for (int j = c; j < m; j++) a[rk][j] *= inv;
        for (int r = 0; r < n; r++) {
            if (r == rk || a[r][c] == 0) continue;
            T fac = a[r][c];
            for (int j = c; j < m; j++)
                a[r][j] -= fac * a[rk][j];
        }
        rk++;
    }
    vector<T> sol(m - 1);
    vector<vector<T>> basis;
    for (int r = rk; r < n; r++)
        if (a[r][m - 1] != 0)
            return {-1, sol, basis};
    for (int c = 0; c < m - 1; c++)
        if (pos[c] != -1)

```

```

        sol[c] = a[pos[c]][m - 1];
    for (int c = 0; c < m - 1; c++)
        if (pos[c] == -1) {
            vector<T> v(m - 1);
            v[c] = 1;
            for (int j = 0; j < m - 1; j++)
                if (pos[j] != -1)
                    v[j] = -a[pos[j]][c];
            basis.push_back(v);
        }
    return {rk, sol, basis};
} // 0e6dd7
template<class T> using Matrix = vector<vector<T>>;

```

6.9 Integer Partition [83bc9d]

```

// CSES_Sum_of_Divisors
const int Mod = 1E9 + 7;
const int inv_2 = 500000004;
// n / 1 * 1 + n / 2 * 2 + n / 3 * 3 + ... + n / n * n
void integerPartition() {
    ll ans = 0, n; cin >> n;
    for (ll l = 1, r; l <= n; l = r + 1) {
        r = n / (n / l);
        ll val = n / l; // n / l 到 n / r 一樣的值
        ll sum = (((l + r) %
            Mod) * ((r - l + 1) % Mod)) % Mod * inv_2; // l 加到 r
        val %= Mod; sum %= Mod;
        ans += val * sum;
        ans %= Mod;
    }
    cout << ans << "\n";
}

```

6.10 Theorems

6.10.1 Basic Theorems

- $a^{m-1} \equiv 1 \pmod{m}$
- $a^{m-2} \equiv a^{-1} \pmod{m}$
- $\tau(N) = \prod_{i=1}^k (p_i^0 + p_i^1 + \dots + p_i^{a_i})$
- $\prod_{d|N} d = N^{\tau(N)}$
- Highly Composite Number:
 - $(10^6): 720720 = 2^4 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11 \cdot 13, \tau = 240$
 - $(10^9): 735134400 = 2^6 \cdot 3^3 \cdot 5^2 \cdot 7 \cdot 11 \cdot 13 \cdot 17, \tau = 1344$
 - $(10^{12}): 963761198400 = 2^6 \cdot 3^4 \cdot 5^2 \cdot 7 \cdot 11 \cdot 13 \cdot 17 \cdot 19 \cdot 23, \tau = 6720$
 - $(10^{15}): 866421317361600 = 2^6 \cdot 3^4 \cdot 5^2 \cdot 7 \cdot 11 \cdot 13 \cdot 17 \cdot 19 \cdot 23 \cdot 29 \cdot 31, \tau = 26880$
 - $(10^{18}): 897612484786617600 = 2^8 \cdot 3^4 \cdot 5^2 \cdot 7^2 \cdot 11 \cdot 13 \cdot 17 \cdot 19 \cdot 23 \cdot 29 \cdot 31 \cdot 37, \tau = 103680$

6.10.2 Combination

- $C(n, m) = C(n, m-1) * (n-m+1) / m$
- $C(n+1, m) = C(n, m) + C(n, m-1)$
- $C(n, k) \equiv 1 \pmod{2} \Leftrightarrow$ all bit of $k \leq$ all bit of n in binary
- $(x+y)^n = \sum_{k=0}^n C(n, k) x^{n-k} y^k$

6.10.3 Game Theorem

- sg 值為 0 代表先手必敗
- 當前 sg 值 = 經過一方操作後，另一方所有可能的後繼狀態的 mex
- 若某個後繼狀態由多個狀態構成，就 xor 起來當作此後繼狀態的 sg 值
- 單組基礎 nim 的 sg 值為本身的原因: $f(0) = 0, f(1) = \text{mex}(f(0)) = 1, f(2) = \text{mex}(f(0), f(1)) = 2, \dots$ 都是自己
- 多組賽局可以把 sg 值 xor 起來，當成最後的 sg 值，nim 也是一樣，且由於 xor 性質，如果可以快速知道 $sg(1)g(2)\dots g(n)$ ，就可以用 xor 性質處理不連續組合

6.10.4 Pisano Period

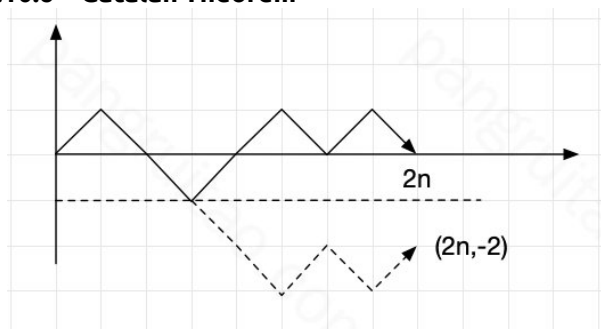
- $\pi(ab) = \text{lcm}(\pi(a), \pi(b)) \quad (\gcd(a, b) = 1)$
- $\pi(p^e) | \pi(p) \cdot p^{e-1}$
- $\pi(p) | p^2 - 1 \quad (p \neq 2, 5)$
- $\pi(2) = 3, \pi(5) = 20$
- so can deal with $p \approx 10^9$ in long long

6.10.5 Burnside's lemma

$$|X/G| = \frac{1}{|G|} \sum_{g \in G} |X^g|$$

- G : 各種翻轉操作所構成的置換群
- X/G : 本質不同的方案的集合
- X^g : 對於某一種操作 g ，所有方案中，經過 g 這種翻轉後保持不變的方案
- 的集合
- 集合取絕對值代表集合數

6.10.6 Catalan Theorem



1. n 個往上 n 個往下，先枚舉所有情況 $\frac{(2n)!}{n!n!} = C_n^{2n}$
2. 扣掉非法的，有多少種可能讓最後的點落在 $(2n, -2)$

假設往上有 x 個，往下有 y 個，會有：

$$\begin{cases} x+y=2n \\ y-x=2 \end{cases} \Rightarrow \begin{cases} x=n-1 \\ y=n+1 \end{cases}$$

所以只要扣掉 C_{n-1}^{2n} 即可

6.10.7 Inversion

- 數論分塊可以快速計算一些含有除法向下取整的和式，就是像 $\sum_{i=1}^n f(i)g(\lfloor \frac{n}{i} \rfloor)$ 的和式。當可以在 $O(1)$ 內計算 $f(r) - f(l)$ 或已經預處理出 f 的前綴和時，數論分塊就可以在 $O(\sqrt{n})$ 的時間內計算上述和式的值。
- 迪利克雷捲積 $h(x) = \sum_{d|x} f(d)g(\frac{x}{d})$
- 積性函數

- $\epsilon(n)$: 只在 $n=1$ 時為 1，其餘情況皆為 0。
- 莫比烏斯函數

1.

$$\sum_{d|n} \mu(d) = \begin{cases} 1 & \text{for } n=1 \\ 0 & \text{for } n \neq 1 \end{cases}$$

2. μ 是常數函數 1 的反元素

$$\Rightarrow \mu * 1 = \epsilon$$

- ϕ 歐拉函數: x 以下與 x 互質的數量

$$\begin{aligned} * \phi(p_1^{a_1} \dots p_k^{a_k}) &= (p_1^{a_1-1}(p_1-1)) \dots (p_k^{a_k-1}(p_k-1)) \\ * n &= \sum_{d|n} \phi(d) \end{aligned}$$

$$\begin{aligned} * \phi * 1 &= \sum_{d|n} \phi(\frac{n}{d}) \quad \text{質因數分解} \\ &= \sum_{i=0}^c \phi(p^i) \\ &= 1 + p^0(p-1) + p^1(p-1) + \dots + p^{c-1}(p-1) \\ &= p^c \\ &= id \end{aligned}$$

- 莫比烏斯反演公式

$$\begin{aligned} - f(n) &= \sum_{d|n} g(d) \Leftrightarrow g(n) = \sum_{d|n} \mu(d) f(\frac{n}{d}) \\ - f(n) &= \sum_{n|d} g(d) \Leftrightarrow g(n) = \sum_{n|d} \mu(\frac{d}{n}) f(d) \end{aligned}$$

- 例子

$$\begin{aligned} &\sum_{i=a}^b \sum_{j=c}^d [gcd(i, j) = k] \\ &\Rightarrow \sum_{i=1}^x \sum_{j=1}^y [gcd(i, j) = k] \\ &= \sum_{i=1}^{\lfloor \frac{x}{k} \rfloor} \sum_{j=1}^{\lfloor \frac{y}{k} \rfloor} \epsilon(gcd(i, j)) \\ &= \sum_{i=1}^{\lfloor \frac{x}{k} \rfloor} \sum_{j=1}^{\lfloor \frac{y}{k} \rfloor} \sum_{d|gcd(i, j)} \mu(d) \\ &= \sum_{d=1}^{\infty} \mu(d) \sum_{i=1}^{\lfloor \frac{x}{k} \rfloor} [d|i] \sum_{j=1}^{\lfloor \frac{y}{k} \rfloor} [d|j] \quad d \text{ 可整除 } i \text{ 時為 } 1 \\ &= \sum_{d=1}^{\min(\lfloor \frac{x}{k} \rfloor, \lfloor \frac{y}{k} \rfloor)} \mu(d) \lfloor \frac{x}{kd} \rfloor \lfloor \frac{y}{kd} \rfloor \end{aligned}$$

6.11 Mobius Inverse [d41d8c]

```
void solve() { // pref: pref of mu
    ll a, b, c, d, k; cin >> a >> b >> c >> d >> k;
    sieve(N);
    auto cal = [&](ll x, ll y) -> int {
        int res = 0;
        for (int l = 1, r; l <= min(x, y); l = r + 1) {
            r = min(x / (x / l), y / (y / l));
            res += (pref[r] - pref[l - 1]) * (x / l) * (y / l);
        }
        return res;
    };
    cout <<
        cal(b / k, d / k) - cal((a - 1) / k, d / k) - cal(b / k, (c - 1) / k) + cal((a - 1) / k, (c - 1) / k) << "\n";
}
```

6.12 LCM Convolution [a64dbb]

```
// ak = sum_{d|k} a[d] (zeta, a \ast 1)
// ck = sum_{d|k} mu[d] * a[k/d] (inv zeta, a \ast mu)
template
<class T> vector<T> zeta(vector<T> a, bool inv = false) {
    int n = a.size() - 1;
    for (int p : primes)
        if (inv) for (int i = n / p; i >= 1; i--) a[i * p] -= a[i];
        else for (int i = 1; i <= n / p; i++) a[i * p] += a[i];
    return a;
} // less implementation
if (inv) for (int i = 1; i <= n; i++)
    for (int j = 2 * i; j <= n; j += i) a[j] -= a[i];
else for (int i = n; i >= 1; i--)
    for (int j = 2 * i; j <= n; j += i) a[j] += a[i];
}
// ck = sum_{lcm(i,j)=k} a[i] * b[j]
template<class T> vector<T> conv(vector<T> a, vector<T> b) {
    a = zeta(a), b = zeta(b);
    for (int i = 1; i < a.size(); i++) a[i] *= b[i];
    return zeta(a, true);
}
```

6.13 exLucas [f8047f]

```
ll legendre(ll n, int p) { // n! 中質數 p 的幂次
    ll res = 0;
    while (n) { n /= p, res += n; }
    return res;
}
Z C_pe(ll n, ll m, int p, int e) {
    int pe = power(p, e);
    Z::setMod(pe); // inv 要用 exgcd
    vector<Z> fac(pe); fac[0] = 1;
    for (int j = 1; j < pe; j++) fac[j] = fac[j - 1] * (j % p ? j : 1);
    Z wilson =
        p == 2 && e >= 3 ? 1 : -1; // (p^e)! = wilson (mod p^e)
    function<Z(ll)> fastFac = [&](ll n)
    { return n ? power
        (wilson, n / pe) * fac[n % pe] * fastFac(n / p) : 1; };
    ll vp = legendre(n, p) - legendre(m, p) - legendre(n - m, p);
    if (vp >= e) return 0;
    return power
        (Z(p), vp) * fastFac(n) / (fastFac(m) * fastFac(n - m));
}
ll exlucas(ll n, ll m, int mod) {
    vector<pair<int, int>> a;
    for (int i = 2; i * i <= mod; i++) {
        if (mod % i) continue;
        int e = 0, p = i;
        while (mod % i == 0) mod /= i, e++, p *= i;
        a.emplace_back(C_pe(n, m, i, e).x, p);
    }
    if (mod != 1) a.emplace_back(C_pe(n, m, mod, 1).x, mod);
    return CRT(a);
}
```

6.14 Quadratic Residue [db1f2f]

```
int jacobi(int x, int p) {
    int s = 1;
    for (; p > 1; ) {
        x %= p;
        if (x == 0) return 0;
        const int r = __builtin_ctz(x);
        if ((r & 1) && ((p + 2) & 4)) s = -s;
        x >>= r;
        if (x & p & 2) s = -s;
        swap(x, p);
    }
    return s;
}
template<class Z> int quadraticResidue(Z x) {
    int p = Z::p();
    if (p == 2) return x.x & 1;
    const int jc = jacobi(x.x, p);
    if (jc == 0) return 0;
    if (jc == -1) return -1;
    Z b, d;
    while (true) {
        b = rand(), d = b * b - x;
        if (jacobi(d.x, p) == -1) break;
    }
    Z f0 = b, f1 = 1, g0 = 1, g1 = 0, tmp;
    for (int e = (p + 1) >> 1; e; e >>= 1) {
        if (e & 1) {
            tmp = g0 * f0 + d * (g1 * f1);
            g1 = g0 * f1 + g1 * f0, g0 = tmp;
        }
        tmp = f0 * f0 + d * (f1 * f1);
        f1 = f0 * f1 * 2, f0 = tmp;
    }
    return min(g0.x, p - g0.x);
}
```

6.15 BSGS [615b88]

```
// a^x = b (mod m)
// x = A * sq - B (0 <= A, B <= sq)
vector<ll> BSGS(int a, int b, int m) { // gcd(a, m) = 1
```

```

Z::setMod(m);
vector<ll> res; // O(sq)
unordered_map<int, int> mp;
ll sq = 1; while (sq * sq < m) sq++;
Z rhs = b;
for (int B = 0; B <= sq; B++) {
    mp[rhs.x] = B;
    rhs *= a;
}
Z a_pow_sq = power(Z(a), sq), lhs = a_pow_sq;
for (int A = 1; A <= sq; A++) {
    if (mp.find(lhs.x) != mp.end()) {
        int B = mp[lhs.x];
        res.push_back(A * sq - B);
    }
}
return res;
} // 468f7b
int exBSGS(int a, int b, int m) {
    if (b == 1 || m == 1) return 0;
    if (!a) return b ? -1 : 1;
    int k = 0;
    ll tmp = 1;
    while (__gcd(a, m) != 1) {
        int d = __gcd(a, m);
        if (b % d != 0) return -1;
        b /= d, m /= d, k++;
        tmp = tmp * (a / d) % m;
        if (tmp == b) return k;
    }
    Z::setMod(m);
    Z b_ = Z(b) * Z(tmp).inv();
    auto xs = BSGS(a, b_.x, m);
    return xs.empty() ? -1 : *min_element(xs.begin(), xs.end()) + k;
} // 3ed312

```

7 Search and Greedy

7.1 Binary Search [d41d8c]

```

void binarySearch() {
    // 二分找上界
    // 如果無解會 = 原 lo, lo 要先 - 1
    while (lo < hi) {
        int x = (lo + hi + 1) / 2;
        if (check(x)) lo = x;
        else hi = x - 1;
    }
    cout << lo;
    // 二分找下界
    // 如果無解會 = 原 hi, hi 要先 + 1
    while (lo < hi) {
        int x = (lo + hi) / 2;
        if (check(m)) hi = x;
        else lo = x + 1;
    }
    cout << lo;
}

```

7.2 Ternary Search [d41d8c]

```

void ternarySearch() {
    int lo = 0, hi = 10;
    while (lo < hi) {
        int xl = lo + (hi - lo) / 3;
        int xr = hi - (hi - lo) / 3;
        int resl = calc(xl), resr = calc(xr);
        if (resl < resr) {
            lo = xl + 1;
        } else {
            hi = xr - 1;
        }
    }
}

```

8 DP

8.1 Projects [2ec0af]

```

// 排程有權重問題(strict), 價值最多且時間最少
void projects(int n, vector<array<int, 4>> a) {
    // a[i] = {t, s, w, idx}
    sort(a.begin(), a.end());
    vector<ll> ts(n + 1); // ensure ts[0] is minimum
    for (int i = 1; i <= n; i++) ts[i] = a[i][0];
    vector<array<ll, 3>> dp(n + 1); // {-w, time, 上個是誰}
    for (int i = 1; i <= n; i++) {
        int id = --lower_bound(all(ts), a[i][1]) - ts.begin();
        dp[i] = {dp[id][0], dp[id][1], -1};
        ll nw = dp[id][0] - a[i][2];
        ll nt = dp[id][1] + a[i][0] - a[i][1];
        chmin(dp[i], {nw, nt, id});
    }
    vector<int> used;
    for (int i = n; i != 0; i--) {
        if (dp[i][2] != -1) {
            used.push_back(a[i][3]);
            i = dp[i][2];
        } else i--;
    }
    reverse(used.begin(), used.end());
}

```

8.2 Bitmask [1a713b]

```

void hamiltonianPath() {
    int n, m; cin >> n >> m;
    vector<vector<int>> adj(n);
    for (int i = 0; i < m; i++) {
        int u, v; cin >> u >> v;
        adj[--u].push_back(--v);
    }
    // 以...為終點, 走過...
    vector dp(n, vector<int>(1 << n));
    dp[0][1] = 1;
    for (int mask = 1; mask < 1 << n; mask++) {
        if ((mask & 1) == 0) continue;
        for (int i = 0; i < n; i++) {
            if ((mask >> i & 1) == 0) continue;
            if (i == n - 1 && mask != (1 << n) - 1) continue;
            int pre = mask ^ (1 << i);
            for (int j : adj[i]) {
                if ((pre >> j & 1) == 0) continue;
                dp[i][mask] = (dp[i][mask] + dp[j][pre]) % Mod;
            }
        }
    }
    cout << dp[n - 1][(1 << n) - 1] << "\n";
}

void minClique() { // 移掉一些邊, 讓整張圖由最少團組成
    int n, m;
    cin >> n >> m;
    vector<int> g(n);
    for (int i = 0; i < m; i++) {
        int u, v;
        cin >> u >> v;
        u--, v--;
        g[u] |= 1 << v, g[v] |= 1 << u;
    }
    vector<int> w(1 << n, 1E9), dp(1 << n, 1E9);
    w[0] = 1, dp[0] = 0;
    for (int mask = 0; mask < 1 << n; mask++) // 先正常 dp
        for (int i = 0; i < n; i++) {
            int pre = mask ^ (1 << i);
            if (w[pre] == 1 && (g[i] & pre) == pre)
                w[mask] = 1; // i 有連到所有 pre
        }
    for (int mask = 0; mask < 1 << n; mask++) // 然後枚舉子集 dp
        for (int sub = mask; sub; --sub &= mask)
            dp[mask] = min(dp[mask], dp[mask ^ sub] + w[sub]);
    cout << dp[(1 << n) - 1] << "\n";
}

```

8.3 Monotonic Queue [c9ba14]

```

// 應用: dp(i) = h(i) + max(A(j)), for l(i) <= j <= r(i)
// A(j) 可能包含 dp(j), h(i) 可 O(1)
void boundedKnapsack() {
    int n, k; // O(nk)
    vector<int> w(n), v(n), num(n);
    deque<int> q;
    // 將同餘的數分在同一組
    // 每次取出連續 num[i] 格中最大值
    // g_x = max_{k=0}^{num[i]} (g_{x-k} + v_i * k)
    // G_x = g_{x} - v_i * x
    // x 代 x-k => v_i * (x-k)
    // g_x = max_{k=0}^{num[i]} (G_{x-k} + v_i * x)
    vector<vector<ll>> dp(2, vector<ll>(k + 1));
    for (int i = 0; i < n; i++) {
        for (int r = 0; r < w[i]; r++) { // 餘數
            q.clear(); // q 記錄在 x = i 時的 dp 有單調性
            for (int x = 0; x * w[i] + r <= k; x++) {
                while (!q.empty() &&
                    q.front() < x - num[i]) q.pop_front(); // 維護遞減
                ll nxt = dp[0][x * w[i] + r] - x * v[i];
                while (!q.empty() && dp[0][q.back()] * w[i] + r - q.back() * v[i] < nxt) q.pop_back();
                q.push_back(x);
                dp[1][x * w[i] + r] = dp[0][q.front() * w[i] + r] - q.front() * v[i] + x * v[i];
            }
            swap(dp[0], dp[1]);
        }
    }
    cout << dp[0][k] << "\n";
}

```

8.4 Digit [1c2af1]

```

// 只含 0, 1, 8 的回文數
const int N = 44;
ll memo[N + 1][N + 1]; // memset -1, once
ll solve(string s) {
    vector<int> a{0};
    while (!s.empty()) {
        a.push_back(s.back() - '0');
        s.pop_back();
    }
    int len = a.size() - 1;
    vector<int> now(len + 1, -1); // 紀錄目前的數字
    return [&](this
        auto &&self, int p, int eff, bool f0, bool lin) -> ll {

```

```

if (!p) return 1; // eff: 扣除前導 0 的長度
if (!lim && !f0 && memo[p][eff] != -1) return memo[p][eff];
ll res = 0;
int lst = lim ? a[p] : 9; // or 1 for binary
for (int i = 0; i <= lst; i++) {
    if (i != 0 && i != 1 && i != 8) continue;
    bool nlim = lim && i == lst;
    now[p] = i;
    if (f0 && i == 0) { // 處理前導零
        res += self(p - 1, eff - 1, true, nlim);
    } else if (p > eff / 2) { // 前半段
        res += self(p - 1, eff, false, nlim);
    } else if (now[p] == now[eff - p + 1]) {
        res += self(p - 1, eff, false, nlim);
    }
}
if (!lim && !f0) memo[p][eff] = res;
return res;
} (len, len, true, true);
}
ll memo[N + 1][10]; // memset -1, once
// 不貼上限、
    不前導零、枚舉 p 往後所有數字中，每個 digit x 出現的次數
ll solve(int x, ll n) {
    vector<int> a{0};
    vector<ll> sum{0}; // 貼上限時，位數 p 以下有多少數字
    while (n) {
        a.push_back(n % 10);
        sum.push_back(
            (sum.back() + (n % 10) * power(10, a.size() - 2));
        );
        n /= 10;
    }
    int len = a.size() - 1;
    return
        [&](this auto &&self, int p, bool f0, bool lim) -> ll {
            if (!p) return 0; // 貢獻來自於當前 digit
            if (!lim && !f0 && memo[p][x] != -1) return memo[p][x];
            ll res = 0;
            int lst = lim ? a[p] : 9; // or 1 for binary
            for (int i = 0; i <= lst; i++) {
                bool nlim = lim && i == lst;
                if (f0 && i == 0) { // 處理前導零
                    res += self(p - 1, true, nlim);
                } else if (nlim) { // 枚舉的前幾位都貼在 n 上
                    res += self(p - 1, false, nlim);
                } else if (i == x) res += sum[p - 1] + 1;
                else {
                    res += self(p - 1, false, nlim);
                }
                if (i == x) res += power(10, p - 1);
            }
            if (!lim && !f0) memo[p][x] = res;
            return res;
        } (len, true, true);
}
// 連續兩位 digit > 1
ll memo[N + 1][10]; // memset -1, once
// 不貼上限、
    不前導零、前一位為 prv 時、枚舉 p 往後所有數字的合法個數
ll solve(ll x) { // 建 a...
    return [&](this
        auto &&self, int p, int prv, bool f0, bool lim) -> ll {
            if (!p) return 1; // 貢獻來自於當前數字數量
            if (!lim && !f0 && memo[p][prv] != -1) return memo[p][prv];
            ll res = 0;
            int lst = lim ? a[p] : 9; // or 1 for binary
            for (int i = 0; i <= lst; i++) {
                if (prv != -1 && abs(i - prv) < 2) continue;
                bool nlim = lim && i == lst;
                if (f0 && i == 0) { // 處理前導零
                    res += self(p - 1, -1, true, nlim);
                } else {
                    res += self(p - 1, i, false, nlim);
                }
            }
            if (!lim && !f0) memo[p][prv] = res;
            return res;
        } (len, -1, true, true);
}

```

8.5 SOS [ec9d5a]

// 使用情況：跟 bit 與(被)包含有關，且 x 在 1E6 左右
// 題目：一數組，問有多少所有數 & 起來為 0 的集合數
// dp[x] 代表包含 x 的 y 個數(比 x 大且 bit 1 全包含 x 的有幾個)
// 答案應該包含在 dp[0] 內，但是有重複元素，所以考慮容斥
// => ans = $\sum_{i=0}^n (-1)^{\text{pop_count}(i)} 2^{\text{dp}[i]-1}$
// => 全
 部為 0 的個數 - 至少一個為 1 的個數 + 至少兩個為 1 的個數
void solve(int n, vector<int> a) {
 int m = __lg(*max_element(a.begin(), a.end())) + 1;
 // 定義 dp[mask] 為 mask 被包含於 a[i] 的 i 個數
 vector<Z> dp(1 << m);
 for (int i = 0; i < n; i++) dp[a[i]] += 1;
 for (int i = 0; i < m; i++) {
 for (int mask = 0; mask < 1 << m; mask++) {
 if (mask >> i & 1) {

```

                int pre = mask ^ (1 << i);
                dp[pre] += dp[mask];
            }
        }
    }
    for (int i = 0; i < 1 << m; i++) {
        dp[i] = power(Z(2), int(dp[i])) - 1;
    }
    // 排容，dp[i]: & 起來為 i 的集合數
    for (int i = 0; i < m; i++) {
        for (int mask = 0; mask < 1 << m; mask++) {
            if (mask >> i & 1) {
                int pre = mask ^ (1 << i);
                dp[pre] -= dp[mask];
            }
        }
    }
}
// x | y = x, 代表包含於 x 的 y 個數，定義為 dp[x][0]
// x & y = x, 代表包含 x 的 y 個數，定義為 dp[x][1]
// x & y
    != 0, 代表至少有一個位元都為 1 的 y 個數，= n - dp[~x][0]
void solve(int n, vector<int> a) {
    int m = __lg(*max_element(a.begin(), a.end())) + 1;
    vector<array<ll, 2>> dp(1 << m);
    for (int i = 0; i < n; i++) dp[a[i]][0]++, dp[a[i]][1]++;
    for (int i = 0; i < m; i++) {
        for (int mask = 0; mask < 1 << m; mask++) {
            if (mask >> i & 1) {
                int pre = mask ^ (1 << i);
                dp[mask][0] += dp[pre][0];
                dp[pre][1] += dp[mask][1];
            }
        }
    }
}

```

8.6 CHT [bdb950]

// 應用：dp(x) = C(x) + min/max(A(i) * x + B(i)), for i < x
// 斜率遞減，求 min，上凸包，自行轉化問題
// ex: m 遞增取 max => max(mx + b) = -min(-m x - b)
struct Line { // x 盡量從 1 開始
 ll m, b;
 Line(ll m = 0, ll b = 0) : m(m), b(b) {}
 ll eval(ll x) { return m * x + b; }
};
struct CHT { // 斜率單調
 deque<Line> hull;
 CHT() {}
 CHT(Line init) { addLine(init); }
 bool frontBad(Line &l1, Line &l2, ll x) {
 return l1.eval(x) >= l2.eval(x);
 }
 bool backBad(Line &l1, Line &l2, Line &l) {
 // 只要
 // l2 跟 l 的 x 交點 <= l1 跟 l 的 x 交點，l2 就用不到了
 return (l1.b
 - l2.b) * (l1.m - l.m) <= (l1.b - l1.b) * (l2.m - l.m);
 }
 void addLine(Line l) {
 if (hull.size() > 0 && hull.back().m == l.m) {
 if (hull.back().b <= l.b) return;
 else hull.pop_back();
 }
 while (hull.size() > 1 && backBad(hull
 [hull.size() - 2], hull.back(), l)) hull.pop_back();
 hull.push_back(l);
 }
 ll query(ll x) { // 查詢沒單調性需要二分搜
 while (hull.size() >
 0 && frontBad(hull[0], hull[1], x)) hull.pop_front();
 return hull[0].eval(x);
 }
};

8.7 DNC [a29240]

// 應用：切 k 段問題，且滿足四邊形不等式
// w(a,c) + w(b,d) <= w(a,d) + w(b,c)
// dp[k][j] = min(dp[k - 1][i] + cost[i][j])
// O(K N logN)
auto getCost = [&](int l, int r) {}; // (l, r], 1-based
vector<ll> dp(n + 1, inf); dp[0] = 0;
for (int j = 0; j < k; j++) {
 vector<ll> ndp(n + 1, inf);
 [&](this auto &&self, int l, int r, int optl, int optr) {
 if (l > r) return;
 int m = (l + r) / 2, opt = 0;
 ndp[m] = inf;
 for (int i = optl; i <= min(m - 1, optr); i++) {
 ll x = dp[i] + getCost(i, m);
 if (x < ndp[m]) ndp[m] = x, opt = i;
 }
 self(l, m - 1, optl, opt);
 self(m + 1, r, opt, optr);
 } (1, n, j, n - 1);
 dp = ndp;
}

8.8 LiChao Segment Tree [2a9325]

```
// 應用: dp(i) = h(i) + min/max(A(j)X(i) + B(j)), for j <= r(i)
//          y = c + m * x + b
template<class T, class F = less<ll>>
struct LiChaoSeg {
    F cmp = F();
    static const T inf = max(numeric_limits<T>::lowest() / 2, numeric_limits<T>::max() / 2, F());
    struct Line {
        T m, b;
        Line(T m = 0, T b = inf) : m(m), b(b) {}
        T eval(T x) const { return m * x + b; }
    };
    struct Node {
        Line line;
        ll l = -1, r = -1;
    };
    ll n;
    vector<Node> nd;
    LiChaoSeg(ll n) : n(n) { newNode(); }
    void addLine(Line line) { update(0, 0, n, line); }
    void rangeAddLine(Line line, ll ql, ll qr) { rangeUpdate(0, 0, n, ql, qr, line); }
    T query(ll x) { return query(x, 0, 0, n); }
private:
    int newNode() {
        nd.emplace_back();
        return nd.size() - 1;
    }
    void update(int p, ll l, ll r, Line line) {
        ll m = (l + r) / 2;
        bool left = cmp(line.eval(l), nd[p].line.eval(l));
        bool mid = cmp(line.eval(m), nd[p].line.eval(m));
        if (mid) swap(nd[p].line, line);
        if (r - l == 1) return;
        if (left != mid) {
            if (nd[p].l == -1) nd[p].l = newNode();
            update(nd[p].l, l, m, line);
        } else {
            if (nd[p].r == -1) nd[p].r = newNode();
            update(nd[p].r, m, r, line);
        }
    }
    void rangeUpdate(int p, ll l, ll r, ll ql, ll qr, Line line) {
        if (r <= ql || l >= qr) return;
        if (ql <= l && r <= qr) return update(p, l, r, line);
        if (nd[p].l == -1) nd[p].l = newNode();
        if (nd[p].r == -1) nd[p].r = newNode();
        ll m = (l + r) / 2;
        rangeUpdate(nd[p].l, l, m, ql, qr, line);
        rangeUpdate(nd[p].r, m, r, ql, qr, line);
    }
    T query(ll x, int p, ll l, ll r) {
        if (p == -1) return inf;
        ll m = (l + r) / 2;
        if (x < m) return min(nd[p].line.eval(x), query(x, nd[p].l, l, m), cmp);
        else return min(nd[p].line.eval(x), query(x, nd[p].r, m, r), cmp);
    }
};
```

9 Geometry

9.1 Basic [54bd6d]

```
const double eps = 1E-9;
template<class T> int sign(T x) {
    return x < -T(eps) ? -1 : x > T(eps) ? 1 : 0;
}
template<class T> struct Pt {
    T x, y; // C++ 20: P(...) -> P[...]
    Pt operator-(const Pt& p) const { return {x - p.x, y - p.y}; }
    Pt operator+(const Pt& p) const { return {x + p.x, y + p.y}; }
    Pt operator-(const Pt& p) const { return {x - p.x, y - p.y}; }
    Pt operator*(const T k) const { return {x * k, y * k}; }
    Pt operator/(const T k) const { return {x / k, y / k}; }
    bool operator<(const Pt& p) const {
        return sign(x - p.x) ? x < p.x : sign(y - p.y) < 0;
    }
    bool operator==(const Pt& p) const { return !sign(x - p.x) && !sign(y - p.y); }
    bool operator!=(const Pt& p) const { return !(*this == p); }
};
using P = Pt<double>; // 0184e8
istream &
operator>>(istream& is, P& p) { return is >> p.x >> p.y; }
ostream &operator<<(ostream& os, const P& p) {
    return os << "(" << p.x << ", " << p.y << ")";
} // 716e78
```

```
struct Line { P a, b; };
template<class T>
T dot(Pt<T> a, Pt<T> b) { return a.x * b.x + a.y * b.y; }
template<class T> T
cross(Pt<T> a, Pt<T> b) { return a.x * b.y - a.y * b.x; }
template<class T> T abs2(Pt<T> p) { return dot(p, p); }
double abs(P p) { return sqrt(abs2(p)); }
double abs(Line l) { return abs(l.a - l.b); }
int dir(P p, Line l) // left -1, right 1, on 0
{ return -sign(cross(l.b - l.a, p - l.a)); }
bool btw(P p, Line l) // c on segment ab?
```

```
{ return dir(p, l) == 0 && sign(dot(p - l.a, p - l.b)) <= 0; }
P norm(P p) { return p / abs(p); }
P rot(P p) { return {-p.y, p.x}; } // 90 degree CCW
P rot(P p, double d) { // CCW, d = degree * PI / 180
    double c = cos(d), s = sin(d);
    return {p.x * c - p.y * s, p.x * s + p.y * c};
} // f6f390
```

```
bool parallel(Line l1, Line l2) {
    return sign(cross(l1.b - l1.a, l2.b - l2.a)) == 0; }
P lineIntersection(Line l1, Line l2) {
    return l1.a + (l1.b - l1.a) * (cross(l2.b - l2.a, l1.a - l2.a) / cross(l2.b - l2.a, l1.a - l1.b)); }
P projvec(P p, Line l) {
    P v = l.b - l.a;
    return l.a + v * (dot(p - l.a, v) / abs2(v));
} // 554901
```

9.2 Sweep And Polar Sort [808d8e]

```
// sort(ps.begin(), ps.end(), SweepCmp(l));
// set<P, SweepCmp> s{SweepCmp(l)};
struct SweepCmp {
    // 以 l 為掃描線, 點從最左邊排到最右邊 (dir: -1 -> 1)
    P d; // 同一條掃描線上的點, 依 l 的方向由起點排到終點
    SweepCmp(Line l) : d(l.b - l.a) {}
    bool operator()(P p, P q) const {
        int c = sign(cross(d, p - q));
        return c != 0 ? c > 0 : sign(dot(d, p - q)) < 0;
    }
}; // 17f9fd
struct PolarCmp {
    // 以 l.a 為中心, 從 l 的方向開始轉, 角度範圍 [0, 2 * PI)
    P o, d;
    int s;
    PolarCmp(Line l, int dir = 0) // dir = 0: CCW, 1: CW
        : o(l.a), d(l.b - l.a), s(dir ? -1 : 1) {}
    int halfP
        p) const { // 中心 -> 0, [0, PI) -> 1, [PI, 2 * PI) -> 2
        int c = sign(cross(d, p)) * s, t = sign(dot(d, p));
        return c == 0 && t == 0 ? 0 : c < 0 || (c == 0 && t < 0) ? 2 : 1;
    }
    bool operator()(P p, P q) const {
        p = p - o, q = q - o;
        if (half(p) != half(q)) return half(p) < half(q);
        int c = sign(cross(p, q)) * s;
        // 1: 同 set 視為相等, pbds_multiset >= (不管順序)
        return c > 0;
        // 2: 同點視為相等, 距離近的排前面, pbds_multiset <=
        return c != 0 ? c > 0 : abs2(p) < abs2(q);
    }
}; // 1: 77c30c, 2: 73c8dc
```

9.3 Convex Hull [3413ae]

```
vector<P> convexHull(vector<P> a) {
    sort(a.begin(), a.end());
    a.resize(unique(a.begin(), a.end()) - a.begin());
    if (a.size() <= 1) return a;
    vector<P> h(a.size() * 2);
    int s = 0, t = 0;
    for (int i = 0; i < 2; i++, s = -t) {
        for (P p : a) {
            while (t >= s + 2 && sign(
                cross(h[t - 1] - h[t - 2], p - h[t - 2])) <= 0) t--;
            h[t++] = p;
        }
        reverse(a.begin(), a.end());
    }
    return {h.begin(), h.begin() + t};
}
```

9.4 Distance [039d8a]

```
double distPL(P p, Line l) {
    return abs(cross(l.a - l.b, l.a - p)) / abs(l);
} // 501648
double distancePS(P p, Line l) {
    if (sign(dot(p - l.a, l.b - l.a)) < 0) return abs(p - l.a);
    if (sign(dot(p - l.b, l.a - l.b)) < 0) return abs(p - l.b);
    return distPL(p, l);
} // ffe672
double distanceSS(Line l1, Line l2) {
    if (get<0>(segmentIntersection(l1, l2)) != 0) return 0.0;
    return min({distancePS(l1.a, l2), distancePS(
        l1.b, l2), distancePS(l2.a, l1), distancePS(l2.b, l1)});
} // fa7723
```

9.5 Point In Polygon [d1a2ce]

```
bool pointInPolygon(P a, const vector<P> &p) {
    int n = p.size(), t = 0;
    for (int i = 0; i < n; i++)
        if (btw(a, {p[i], p[(i + 1) % n]})) return true;
    for (int i = 0; i < n; i++) {
        P u = p[i], v = p[(i + 1) % n];
        if (u.x < a.x && v.x >= a.x && dir(a, {v, u}) < 0) t ^= 1;
        if (u.x >= a.x && v.x < a.x && dir(a, {u, v}) < 0) t ^= 1;
    }
    return t == 1;
}
```

9.6 Point In Convex Polygon [3bf18c]

```
// 0 : strictly outside
// 1 : on boundary
// 2 : strictly inside
int pointInConvexPolygon(P a, const vector<P> &p) {
    int n = p.size();
    if (n == 0) return 0;
    else if (n <= 2) return btw(a, {p[0], p.back()});
    if (btw
        (a, {p[0], p[1]}) || btw(a, {p[0], p[n - 1]})) return 1;
    else if (dir(a, {p[1],
        p[0]}) < 0 || dir(a, {p[0], p[n - 1]}) < 0) return 0;
    int lo = 1, hi = n - 2;
    while (lo < hi) {
        int x = (lo + hi + 1) / 2;
        if (dir(a, {p[0], p[x]}) < 0) lo = x;
        else hi = x - 1;
    }
    if (dir(a, {p[lo], p[lo + 1]}) < 0) return 2;
    else return btw(a, {p[lo], p[lo + 1]});
}
```

9.7 Line Intersects Polygon [254956]

```
bool lineIntersectsPolygon(Line l, const vector<P> &p) {
    int n = p.size();
    for (int i = 0; i < n; i++) {
        P a = p[i], b = p[(i + 1) % n];
        if (dir(a, l) == 0 || dir(b, l) == 0) return true;
        if ((dir(a, l) < 0) ^ (dir(b, l) < 0)) return true;
    }
    return false;
}
```

9.8 Segment Intersection [ff1cf8]

```
// 0 : not intersect
// 1 : strictly intersect
// 2 : overlap
// 3 : intersect at endpoint
tuple<int, P, P> segmentIntersection(Line l1, Line l2) {
    if (parallel(l1, l2)) {
        if (dir(l1.a, l2) != 0) return {0, {}, {}};
        else {
            vector<P> res;
            if (btw(l1.a, l2)) res.push_back(l1.a);
            if (btw(l1.b, l2)) res.push_back(l1.b);
            if (btw(l2.a, l1)) res.push_back(l2.a);
            if (btw(l2.b, l1)) res.push_back(l2.b);
            if (res.empty()) return {0, {}, {}};
            sort(res.begin(), res.end());
            res.resize(unique(res.begin(), res.end()) - res.begin());
            if (res.size() == 1) return {3, res[0], res[0]};
            else return {2, res.front(), res.back()};
        }
    }
    auto cp1 = dir(l2.b, l1), cp2 =
        dir(l2.a, l1), cp3 = dir(l1.b, l2), cp4 = dir(l1.a, l2);
    if ((cp1 > 0 && cp2 > 0) || (cp1 < 0 && cp2 < 0) ||
        (cp3 > 0 &&
            cp4 > 0) || (cp3 < 0 && cp4 < 0)) return {0, {}, {}};
    P p = lineIntersection(l1, l2);
    if (cp1 !=
        0 && cp2 != 0 && cp3 != 0 && cp4 != 0) return {1, p, p};
    else return {3, p, p};
}
```

9.9 Segment In Polygon [e28323]

```
bool segmentInPolygon(Line l, const vector<P> &p) {
    int n = p.size();
    vector<P> a{l.a, l.b};
    for (int i = 0; i < n; i++) {
        auto [t, p1,
            p2] = segmentIntersection(l, {p[i], p[(i + 1) % n]});
        if (t == 1 || t == 3) a.push_back(p1);
        else if (t == 2) a.push_back(p1), a.push_back(p2);
    }
    sort(a.begin(), a.end());
    for (int i = 0; i + 1 < a.size(); i++) {
        if (sign(abs(a[i] - a[i + 1])) == 0) continue;
        if (!pointInPolygon
            ((a[i] + a[i + 1]) / 2, p)) return false;
    }
    return true;
}
```

9.10 Half Plane Intersection [24a100]

```
// non-strict
-> >= 都改 >, assert 拿掉, 在外面特判直線(area == 0)
vector<P> hp(vector<Line> lines) {
    auto sgn = [](P p) {
        return p.y > 0 || (p.y == 0 && p.x > 0) ? 1 : -1;
    };
    sort(lines.begin(), lines.end(), [&](auto l1, auto l2) {
        auto d1 = l1.b - l1.a;
        auto d2 = l2.b - l2.a;
        if (sgn(d1) != sgn(d2)) return sgn(d1) == 1;
        return cross(d1, d2) > 0;
    });
    deque<Line> ls;
```

```
deque<P> ps;
for (auto l : lines) {
    if (ls.empty()) {
        ls.push_back(l);
        continue;
    }
    while (!ps.empty() &&
        dir(ps.back(), l) >= 0) ps.pop_back(), ls.pop_back();
    while (!ps.empty()
        && dir(ps[0], l) >= 0) ps.pop_front(), ls.pop_front();
    if (sign(cross(
        l.b - l.a, ls.back().b - ls.back().a)) == 0) { // 平行
        if (sign(dot(
            l.b - l.a, ls.back().b - ls.back().a)) > 0) { // 同向
            if (dir(ls.back().a, l) >= 0) { // l 在內側
                assert(ls.size() == 1);
                ls[0] = l;
            }
            continue;
        }
        return {};
    }
    ps.push_back(lineIntersection(ls.back(), l));
    ls.push_back(l);
}
while (!ps.empty() && dir(ps.back(), ls[0]) >= 0)
    ps.pop_back(), ls.pop_back();
if (ls.size() <= 2) return {};
ps.push_back(lineIntersection(ls[0], ls.back()));
return vector(ps.begin(), ps.end());
}
```

9.11 Min Euclidean Distance [723df0]

```
void solve(int n, vector<P> &a) { // recursive solution
    sort(a.begin(), a.end());
    vector<P> t(n);
    cout << [&](this auto &&self, int l, int r) -> ll {
        if (l == r) return 9E18;
        int p = 0, m = (l + r) / 2;
        ll mid = a[m].x;
        ll ans = min(self(l, m), self(m + 1, r));
        merge(a.begin() + l, a.begin() + m + 1,
            a.begin() + m + 1, a.begin() + r + 1,
            t.begin(),
            [](const P &a, const P &b) { return a.y < b.y; });
        copy(t.begin(), t.begin() + r - l + 1, a.begin() + l);
        for (int i = l; i <= r; i++)
            if ((mid
                - a[i].x) * (mid - a[i].x) <= ans) t[p++] = a[i];
        for (int i = 0; i < p; i++) {
            for (int j = i + 1; j < p; j++) {
                if ((t[i].y - t[j].y) * (t[i].y - t[j].y) > ans) break;
                ans = min(ans, abs2(t[i] - t[j]));
            }
        }
        return ans;
    } (0, n - 1) << "\n";
}

// K-D tree solution
struct Info {
    static constexpr int DIM = 2;
    array<ll, DIM> x, L, R;
    ll distl, distr;
    ll f(const Info &i) {
        ll ret = 0;
        if (i.
            L[0] > x[0]) ret += (i.L[0] - x[0]) * (i.L[0] - x[0]);
        if (i.
            R[0] < x[0]) ret += (x[0] - i.R[0]) * (x[0] - i.R[0]);
        if (i.
            L[1] > x[1]) ret += (i.L[1] - x[1]) * (i.L[1] - x[1]);
        if (i.
            R[1] < x[1]) ret += (x[1] - i.R[1]) * (x[1] - i.R[1]);
        return ret;
    }
};
void pull(const Info &l, const Info &r) {
    distl = f(l), distr = f(r);
}

struct KDTree { // 1-indexed
    static constexpr int DIM = Info::DIM;
    int n, rt;
    vector<Info> info;
    vector<int> l, r;
    KDTree(const vector<Info> &
        info) : n(info.size()), info(info), l(n + 1), r(n + 1) {
        rt = build(1, n);
    }
    void pull(int p) {
        info[p].L = info[p].R = info[p].x;
        info[p].pull(info[l[p]], info[r[p]]);
        for (int ch : {l[p], r[p]}) {
            if (!ch) continue;
            for (int k = 0; k < DIM; k++) {
                info[p].L[k] = min(info[p].L[k], info[ch].L[k]);
                info[p].R[k] = max(info[p].R[k], info[ch].R[k]);
            }
        }
    }
    int build(int l, int r) {
        if (r == l) return 0;
    }
};
```

9.12 Max Euclidean Distance [7e3b52]

9.13 Min Circle Cover [96a4b7]

9.14 Min Rectangle Cover [3a0f29]

```

auto get = [&](P p, Line l) -> double {
    return abs(cross(l.a - l.b, l.a - p));
}; // line 到 p 圍成的四邊形面積
int n = p.size(), j = 2, l = 1, r = 1;
p.push_back(p.front());
double ans = 8E18;
vector<P> ps;
for (int i = 0; i < n; i++) {
    while (get(p[j], {p[i],
        p[i + 1]}) <= get(p[(j + 1) % n], {p[i], p[i + 1]}))
        j = (j + 1) % n;
    while (dot(p[i + 1] - p[i], p[r] -
        p[i]) <= dot(p[i + 1] - p[i], p[(r + 1) % n] - p[i]))
        r = (r + 1) % n;
    if (i == 0) l = j;
    while (dot(p[i + 1] - p[i], p[l] -
        p[i]) >= dot(p[i + 1] - p[i], p[(l + 1) % n] - p[i]))
        l = (l + 1) % n;
    double area = get(p[j], {p[i], p[i + 1]});
    double w = dot(p[i] - p[i +
        1], p[l] - p[i]) + dot(p[i + 1] - p[i], p[r] - p[i]);
    area *= w / abs2(p[i + 1] - p[i]);
    if (area < ans) {
        ps.clear(), ans = area;
        Line l1{p[i], p[i + 1]};
        for (auto u : {p[r], p[j], p[l], p[i]}) {
            if (u == l1.b) {
                ps.push_back(u);
                l1 = {u, u + rot(l1.b - l1.a)};
            } else {
                Line l2 = {u, u + rot(l1.b - l1.a)};
                P res = lineIntersection(l1, l2);
                ps.push_back(res);
                l1 = {res, u};
            }
        }
    }
}
return {ans, ps};

```

```
// CCW needed, O(n + m)
double convexPolyDist(vector<P> a, vector<P> b) {
    int n = a.size(), m = b.size();
    int sa = 0, sb = 0;
    for (int i = 1; i < n; i++) if (a[i].y < a[sa].y) sa = i;
    for (int i = 1; i < m; i++) if (b[i].y > b[sb].y) sb = i;
    double ans = 1E18;
    for (int i = 0; i < n; i++) {
        P na = a[sa], nb = a[(sa + 1) % n];
        P ea = nb - na;
        while (sign(cross(ea, b[(sb + 1) % m] - b[sb])) > 0)
            sb = (sb + 1) % m;
        chmin(ans, distanceSS({na, nb}, {b[sb], b[(sb + 1) % m]}));
        sa = (sa + 1) % n;
    }
    return ans;
}
```

```
// (interior, boundary)
pair<ll, ll> latticePoints(int n, vector<P> p, ll area2) {
    ll bnd = 0;
    p.push_back(p[0]);
    for (int i = 0; i < n; i++)
        bnd += gcd
            (abs(p[i + 1].x - p[i].x), abs(p[i + 1].y - p[i].y));
    return {(area2 - bnd) / 2 + 1, bnd};
}
```

```
double polygonUnion(vector<vector<P>> ps) { // CCW needed
    int n = ps.size();
    for (auto &v : ps) v.push_back(v[0]);
    double res = 0;
    auto seg = [(P o, P a, P b) -> double {
        if (b.x - a.x == 0) return (o.y - a.y) / (b.y - a.y);
        return (o.x - a.x) / (b.x - a.x);
    }];
    for (int pi = 0; pi < n; pi++) {
        for (int i = 0; i + 1 < ps[pi].size(); i++) {
            vector<pair<double, int>> e{{0, 0}, {1, 0}};
            for (int pj = 0; pj < n; pj++) {
                if (pi == pj) continue;
                for (int j = 0; j + 1 < ps[pj].size(); j++) {
                    auto c1 = cross(ps
                        [pi][i + 1] - ps[pi][i], ps[pj][j] - ps[pi][i]);
                    auto c2 = cross(ps[pi][
                        i + 1] - ps[pi][i], ps[pj][j + 1] - ps[pi][i]);
                    if (c1 == 0 && c2 == 0) {
                        if (dot(ps[pi][i + 1] - ps[pi][i], ps[pj]
                            [j + 1] - ps[pj][j]) > 0 && (pi - pj) > 0) {
                            e.emplace_back(seg
                                (ps[pj][j], ps[pi][i], ps[pi][i + 1]), 1);
                            e.emplace_back(seg(ps
                                [pj][j + 1], ps[pi][i], ps[pi][i + 1]), -1);
                        }
                    } else {
```

```

        auto s1 = cross(ps[pj]
            ][j + 1] - ps[pj][j], ps[pi][i] - ps[pj][j]);
        auto s2 = cross(ps[pj][j]
            + 1] - ps[pj][j], ps[pi][i + 1] - ps[pj][j]);
        if (c1 >= 0
            && c2 < 0) e.emplace_back(s1 / (s1 - s2), 1);
        else if (c1 < 0 &&
            c2 >= 0) e.emplace_back(s1 / (s1 - s2), -1);
    }
}
sort(e.begin(), e.end());
double pre = clamp(e[0].first, 0.0, 1.0), sum = 0;
int cov = e[0].second;
for (int j = 1; j < e.size(); j++) {
    double now = clamp(e[j].first, 0.0, 1.0);
    if (!cov) sum += now - pre;
    cov += e[j].second;
    pre = now;
}
res += cross(ps[pi][i], ps[pi][i + 1]) * sum;
}
return res / 2;
}

```

10 Polynomial

10.1 FFT [569b66]

```

const double PI = acos(-1.0);
using cd = complex<double>;
void fft(vector<cd> &a, bool inv = false) {
    static vector<cd> w{0, 1};
    int n = a.size();
    for (int k = w.size(); k < n; k *= 2) {
        w.resize(2 * k);
        for (int
            j = 0; j < k; j++) w[k + j] = polar(1.0, PI * j / k);
    }
    if (!inv) {
        for (int k = n / 2; k >= 1; k /= 2)
            for (int i = 0; i < n; i += 2 * k)
                for (int j = 0; j < k; j++) {
                    cd u = a[i + j], v = a[i + j + k];
                    a[i + j] = u + v;
                    a[i + j + k] = (u - v) * w[k + j];
                }
        return;
    }
    for (int k = 1; k < n; k *= 2)
        for (int i = 0; i < n; i += 2 * k)
            for (int j = 0; j < k; j++) {
                cd u = a[i + j], v = a[i + j + k] * w[k + j];
                a[i + j] = u + v;
                a[i + j + k] = u - v;
            }
    reverse(a.begin() + 1, a.end());
    for (auto &x : a) x /= n;
}
// i + j = k, res[1] = a[1] * b[0] + a[0] * b[1]
// reverse b for i - j = k, res[k + b.size() - 1]
template
<class T> vector<double> conv(vector<T> a, vector<T> b) {
    vector<cd> fa(a.begin(), a.end()), fb(b.begin(), b.end());
    int n = 1, tot = a.size() + b.size() - 1;
    while (n < tot) n <= 1;
    fa.resize(n), fb.resize(n);
    fft(fa), fft(fb);
    for (int i = 0; i < n; i++) fa[i] = fa[i] * fb[i];
    fft(fa, true);
    vector<double> res(tot);
    for (int i = 0; i < tot; i++) res[i] = fa[i].real();
    return res; // use lround to int
}

```

10.2 NTT [0eaf84]

```

template<class Z>
void ntt(vector<Z> &a, bool inv = false, const int G = 3) {
    static vector<Z> w{0, 1};
    int n = a.size();
    if (w.size() < n) {
        int k = __builtin_ctz(w.size());
        w.resize(n);
        while ((1 << k) < n) {
            Z u = power(Z(G), (Z::p() - 1) >> (k + 1));
            for (int i = 1 << (k - 1); i < 1 << k; i++) {
                w[i * 2] = w[i];
                w[i * 2 + 1] = w[i] * u;
            }
            k++;
        }
    }
    if (inv) {
        for (int k = 1; k < n; k *= 2) {
            for (int i = 0; i < n; i += 2 * k) {
                for (int j = 0; j < k; j++) {
                    Z u = a[i + j];
                    Z v = a[i + j + k] * w[k + j];
                    a[i + j] = u + v;
                    a[i + j + k] = u - v;
                }
            }
        }
    }
}

```

```

    }
}
reverse(a.begin() + 1, a.end());
Z inv = Z(n).inv();
for (auto &x : a) x *= inv;
} else {
    for (int k = n / 2; k >= 1; k /= 2) {
        for (int i = 0; i < n; i += 2 * k) {
            for (int j = 0; j < k; j++) {
                Z u = a[i + j];
                Z v = a[i + j + k];
                a[i + j] = u + v;
                a[i + j + k] = (u - v) * w[k + j];
            }
        }
    }
}
} // d2e699
template<class Z>
vector<Z> conv(vector<Z> a, vector<Z> b, const int G = 3) {
    if (a.empty() || b.empty()) return {};
    int n = 1, tot = a.size() + b.size() - 1;
    while (n < tot) n <= 1;
    a.resize(n), b.resize(n);
    ntt(a, false, G), ntt(b, false, G);
    for (int i = 0; i < n; i++) a[i] = a[i] * b[i];
    ntt(a, true, G);
    a.resize(tot);
    return a;
} // 7866d6
const int G = 3;
const int P1 = 167772161, P2 = 104857601, P3 = 998244353;
using Z1 = Mint<P1>;
using Z2 = Mint<P2>;
using Z3 = Mint<P3>;
const int I12 = Z2(P1).inv().x;
const int I123 = Z3(1LL * P1 * P2).inv().x;
template<class T> T garner(int r1, int r2, int r3) {
    Z2 t1 = Z2(r2 - r1) * I12;
    Z3 t2 = Z3(r3 - r1 - 1LL * P1 * t1.x) * I123;
    return T(r1) + T(P1) * t1.x + T(P1) * P2 * t2.x;
} // 6def25
template<class Z>
vector<Z> convArb(vector<Z> a, vector<Z> b) {
    vector<Z1> a1(a.size()), b1(b.size());
    vector<Z2> a2(a.size()), b2(b.size());
    vector<Z3> a3(a.size()), b3(b.size());
    for (int i = 0; i < a.size(); i++) {
        a1[i] = a[i].x;
        a2[i] = a[i].x;
        a3[i] = a[i].x;
    }
    for (int i = 0; i < b.size(); i++) {
        b1[i] = b[i].x;
        b2[i] = b[i].x;
        b3[i] = b[i].x;
    }
    auto c1 = conv(a1, b1);
    auto c2 = conv(a2, b2);
    auto c3 = conv(a3, b3);
    int tot = a.size() + b.size() - 1;
    vector<Z> res(tot);
    for (int i = 0; i < tot; i++)
        res[i] = garner<Z>(c1[i].x, c2[i].x, c3[i].x);
    return res;
} // 7e3465

```

10.3 FWT [c53534]

```

template<class T> void fwt(vector<T> &a, bool inv = false) {
    // Or : [[1, 0], [1, 1]] <-> [[1, 0], [-1, 1]]
    // And: [[1, 1], [0, 1]] <-> [[1, -1], [0, 1]]
    // Xor: [[1, 1], [1, -1]] <-> [[0.5, 0.5], [0.5, -0.5]]
    // *inv_2 instead of /2 for Z
    int n = __lg(int(a.size()));
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < 1 << i; j++) {
            if (j >> i & 1) { // And
                T x = a[j ^ (1 << i)];
                T y = a[j];
                if (!inv) {
                    a[j ^ (1 << i)] = x + y;
                    a[j] = y;
                } else {
                    a[j ^ (1 << i)] = x - y;
                    a[j] = y;
                }
            }
        }
    }
}

```

10.4 Polynomial [f7c42d]

```

struct Poly : public vector<Mint<P>> {
    using Z = Mint<P>;
    using vector::vector;
    Poly(const vector<Z> &a) : vector<Z>(a) {}
    Poly operator-() const {
        vector<Z> res(this->size());
        for (int
            i = 0; i < int(res.size()); i++) res[i] = -(*this)[i];
    }
}

```

```

    return Poly(res);
}
friend Poly operator+(Poly a, Poly b) {
    a.resize(max(a.size(), b.size()));
    for (int i = 0; i < b.size(); i++) a[i] += b[i];
    return a;
}
friend Poly operator-(Poly a, Poly b) {
    a.resize(max(a.size(), b.size()));
    for (int i = 0; i < b.size(); i++) a[i] -= b[i];
    return a;
}
friend Poly operator*(Poly a, Poly b) {
    return conv(a, b, G);
}
friend Poly operator*(Poly a, Z x) {
    for (int i = 0; i < a.size(); i++) a[i] *= x;
    return a;
}
friend Poly operator/(Poly a, Z x) {
    for (int i = 0; i < a.size(); i++) a[i] /= x;
    return a;
}
Poly &operator+=(Poly a) { return *this = *this + a; }
Poly &operator-=(Poly a) { return *this = *this - a; }
Poly &operator*=(Poly a) { return *this = *this * a; }
Poly &operator*=(Z a) { return *this = *this * a; }
Poly &operator/=(Z a) { return *this = *this / a; }
// 53f01c
Poly shift(int k) const { // 乘 x^k, k < 0 時捨去低次項
    if (k >= 0) {
        auto b = *this;
        b.insert(b.begin(), k, 0);
        return b;
    } else if (this->size() <= -k) {
        return Poly();
    } else {
        return Poly(this->begin() + (-k), this->end());
    }
} // e6ab95
Poly trunc(int k) const {
    Poly f = *this; f.resize(k); return f;
} // 217dbc
Poly deriv() const {
    if (this->empty()) return Poly();
    Poly res(this->size() - 1);
    for (int i = 0; i < this->size() - 1; i++)
        res[i] = (*this)[i + 1] * (i + 1);
    return res;
} // cceaf0
Poly integr() const {
    Poly res(this->size() + 1);
    for (int i = 0; i < this->size(); i++)
        res[i + 1] = (*this)[i] / (i + 1);
    return res;
} // f86ffe
// 以下帶 m 的都是 mod x^m, 結果只留前 m 項
Poly inv(int m) const { // 1 / f, need f[0] != 0
    Poly x{(*this)[0].inv()};
    int k = 1;
    while (k < m) {
        k *= 2;
        x = (x * (Poly{2} - trunc(k) * x)).trunc(k);
    }
    return x.trunc(m);
} // 8287e6
Poly log(int m) const { // need f[0] = 1
    return (deriv() * inv(m)).integr().trunc(m);
} // bfd979
Poly pow(ll k, int m) const { // f^k, f[0] 可為 0
    if (k == 0) { Poly res(m); res[0] = 1; return res; }
    int i = 0;
    while (i < this->size() && (*this)[i].x == 0) i++;
    if (i == this->size()) || i > 0 && k > (m - 1) / i return Poly(m);
    Z v = (*this)[i];
    auto f = shift(-i) * v.inv();
    return (f.log(m - i * k) * Z(k)).exp(m - i * k).shift(i * k) * power(v, k);
} // 23cb76
Poly sqrt(int m) const { // need quadraticResidue, 無解回傳空
    int k = 0;
    while (k
        < this->size() && (*this)[k].x == 0) k++; // 找前導零
    if (k == this->size()) return Poly(m); // 全零多項式
    if (k % 2 != 0) return Poly(); // 無解: 最低次項為奇數
    int s = quadraticResidue((*this)[k]);
    if (s == -1) return Poly(); // 無解: 係數無平方根
    int ofst = k / 2, r = m - ofst;
    if (r <= 0) return Poly(m);
    Poly h = this->shift(-k) * (*this)[k].inv(), g{1};
    for (int i = 1; i < r; i <= 1) {
        int len = i <= 1;
        g = (g + h.trunc(len) * g.inv(len)).trunc(len) / 2;
    }
    g.resize(r);
    g = (g * Z(s)).shift(ofst).trunc(m);
    return g;
} // 1a3aea
Poly exp(int m) const { // need f[0] = 0

```

```

    Poly x{1};
    int k = 1;
    while (k < m) {
        k *= 2;
        x = (x * (Poly{1} - x.log(k) + trunc(k))).trunc(k);
    }
    return x.trunc(m);
} // 78baba
// 轉置乘法
, res[i] = sum_j f[i + j] * b[j], res.size() = f.size()
Poly mult(Poly b) const {
    if (b.empty()) return Poly();
    int n = b.size();
    reverse(b.begin(), b.end());
    return ((*this) * b).shift(-(n - 1));
} // f7c257
// 多點求值, 回傳 f(x[0]), f(x[1]), ..., 0(n log^2 n)
vector<Z> eval(vector<Z> x) const {
    if (this->size() == 0) return vector<Z>(x.size(), 0);
    const int n = max(x.size(), this->size());
    vector<Poly> q(4 * n);
    vector<Z> ans(x.size());
    x.resize(n);
    function<void>
        (int, int, int)> build = [&](int p, int l, int r) {
        if (r - l == 1) {
            q[p] = Poly{1, -x[l]};
        } else {
            int m = (l + r) / 2;
            build(2 * p, l, m);
            build(2 * p + 1, m, r);
            q[p] = q[2 * p] * q[2 * p + 1];
        }
    };
    build(1, 0, n);
    function<void>(int, int, int, const Poly
        &)> work = [&](int p, int l, int r, const Poly &num) {
        if (r - l == 1) {
            if (l < int(ans.size())) ans[l] = num[0];
        } else {
            int m = (l + r) / 2;
            work(2 * p, l, m, num.mult(q[2 * p + 1]).trunc(m - l));
            work(2 * p + 1, m, r, num.mult(q[2 * p]).trunc(r - m));
        }
    };
    work(1, 0, n, mult(q[1].inv(n)));
    return ans;
} // 9698d5
};

```

10.5 Barlekamp Massey [d6813f]

```

// find s[i] = c[0]
* s[i - 1] + c[1] * s[i - 2] + ... + c[k - 1] * s[i - k]
// return: c(x) = 1 - c_0x - c_1x^2 - ... - c_{k-1}x^k
Poly berlekampMassey(const Poly &s) { // 0(n^2)
    Poly c, oldC;
    int f = -1;
    for (int i = 0; i < s.size(); i++) {
        auto delta = s[i];
        for (int j = 1; j <= c.size(); j++)
            delta -= c[j - 1] * s[i - j];
        if (delta.x == 0) continue;
        if (f == -1) {
            c.resize(i + 1);
            f = i;
        } else {
            auto d = oldC;
            d *= -1;
            d.insert(d.begin(), 1);
            Z df1 = 0;
            for (int j = 1; j <= d.size(); j++)
                df1 += d[j - 1] * s[f + 1 - j];
            assert(df1.x != 0);
            auto coef = delta / df1;
            Poly zeros(i - f - 1);
            zeros.insert(zeros.end(), d.begin(), d.end());
            d = zeros;
            auto temp = c;
            c += d;
            if (i - temp.size() > f - oldC.size()) {
                oldC = temp;
                f = i;
            }
        }
    }
    c *= -1;
    c.insert(c.begin(), 1);
    return c;
}

```

10.6 Linear Recurrence [81be8a]

```

// a_i = c[0] *
a_{i-1} + c[1] * a_{i-2} + ... + c[k-1] * a_{i-k}
// q(x) = 1 - c[0] * x - c[1] * x^2 - ... - c[k-1] * x^k
// p(x) = (a * q).trunc(c.size())
// return: a_n
Z linearRecurrence(Poly p, Poly q, ll n) { // 0(m log m log n)
    int m = q.size() - 1;
    while (n > 0) {

```

```

    auto nq = q;
    for (int i = 1; i <= m; i += 2) nq[i] *= -1;
    auto np = p * nq;
    nq = q * nq;
    for (int i = 0; i < m; i++) p[i] = np[i * 2 + n % 2];
    for (int i = 0; i <= m; i++) q[i] = nq[i * 2];
    n /= 2;
}
return p[0] / q[0];
}

```

Prime	Root	Prime	Root
7681	17	167772161	3
12289	11	104857601	3
40961	3	985661441	3
65537	3	998244353	3
786433	10	1107296257	10
5767169	3	2013265921	31
7340033	3	2810183681	11
23068673	3	2885681153	3
469762049	3	605028353	3
2061584302081	7	1945555039024054273	5
2748779069441	3	9223372036737335297	3

11 Else

11.1 Segment Manipulation [30596f]

```

auto era = [&](set<pair<int, int>> &st, int L, int R) {
    auto it = st.lower_bound({L, -1});
    if (it != st.begin() && prev(it)->second >= L) --it;
    while (it != st.end() && it->first <= R) {
        auto [l, r] = *it;
        it = st.erase(it);
        if (l < L) st.emplace(l, L - 1);
        if (R < r) st.emplace(R + 1, r);
    }
};
// remove + 1 for strictly overlap
auto ins = [&](set<pair<int, int>> &st, int L, int R) {
    auto it = st.lower_bound({L, -1});
    if (it != st.begin() && prev(it)->second + 1 >= L) --it;
    while (it != st.end() && it->first <= R + 1) {
        L = min(L, it->first);
        R = max(R, it->second);
        it = st.erase(it);
    }
    st.emplace(L, R);
};
auto find = [&](const set<pair<int, int>> &st, int i) {
    auto it = st.lower_bound({i + 1, -1});
    if (it != st.begin()) {
        --it;
        if (it->second >= i) return it;
    }
    return st.end();
};

```

11.2 Python [7c66a4]

```

from decimal import * # 高精度浮點數
from fractions import * # 分數
from random import *
from math import *
# set decimal prec bigger if it could overflow in precision
setcontext(
    Context(prec=10, Emax=MAX_EMAX, rounding=ROUND_FLOOR))
# read and print
x = int(input())
a, b, c = list(map(Fraction, input().split()))
arr = list(map(Decimal, input().split()))
print(*arr)
# set
st = set(); st.add((a, b)); st.remove((a, b))
if not (a, b) in st:
# dict
d = dict(); d[(a, b)] = 1; del d[(a, b)]
for (a, b) in d.items():
# random
arr = [randint(1, r) for i in range(size)]
choice([8, 6, 4, 1]) # random pick one
shuffle(arr)

```

11.3 Bigint [024a84]

```

struct Bigint { // not support hex division
private:
    static const int digit = 9; // hex: 7
    static const int base = 10; // hex: 16
    static const int B = power(10, digit);
    Bigint(vector<int> x, int sgn) : x(x), sgn(sgn) {}
    template<class U> vector<int> norm(vector<U> a) {
        if (a.empty()) return {};
        for (int i = 0; i < a.size(); i++) {
            U c = a[i];
            a[i] %= B;
            if (c / B)
                if (i == a.size() - 1) a.push_back(c);
                else a[i + 1] += c;
        }
    }

```

```

    while (a.size() > 1 && a.back() == 0) a.pop_back();
    return {a.begin(), a.end()};
}
void resign() { sgn = x.back() == 0 ? 1 : sgn; }
int toInt(char c) const {
    if (isdigit(c)) return c - '0';
    else return c - 'A' + 10;
}
char toChar(int c) const {
    if (c < 10) return c + '0';
    else return c - 10 + 'A';
}
public:
    int sgn = 1;
    vector<int> x; // 反著存
    Bigint() : x{0}, sgn(1) {}
    Bigint(ll a) { *this = Bigint(std::to_string(a)); }
    Bigint(string s) {
        if (s.empty()) { *this = Bigint(); return; }
        if (s[0] == '-') s.erase(s.begin()), sgn = -1;
        int add = 0, cnt = 0, b = 1;
        while (s.size()) {
            if (cnt == digit) x.push_back(add), add = cnt = 0, b = 1;
            add += toInt(s.back()) * b;
            cnt++, b *= base;
            s.pop_back();
        }
        if (add) x.push_back(add);
        x = norm(x);
    }
    int size() const { return x.size(); }
    Bigint abs() const { return Bigint(x, 1); }
    string to_string() const {
        string res;
        for (int i = 0; i < x.size(); i++) {
            string add; int v = x[i];
            for (int j = 0; j < digit; j++)
                add += toChar(v % base), v /= base;
            res += add;
        }
        while (res.size() > 1 && res.back() == '0') res.pop_back();
        if (sgn == -1) res += '-';
        reverse(res.begin(), res.end());
        return res;
    }
    Bigint operator-() const { return Bigint(x, -sgn); }
    friend Bigint operator+(Bigint a, Bigint b) {
        if (a.sgn != b.sgn) return a - (-b);
        int n = max(a.size(), b.size());
        a.x.resize(n), b.x.resize(n);
        for (int i = 0; i < n; i++) a.x[i] += b.x[i];
        a.x = a.norm(a.x), a.resign();
        return a;
    }
    friend Bigint operator-(Bigint a, Bigint b) {
        if (a.sgn != b.sgn) return a + (-b);
        if (a.abs() < b.abs()) return -(b - a);
        int n = max(a.size(), b.size());
        a.x.resize(n), b.x.resize(n);
        for (int i = 0; i < n; i++) {
            a.x[i] -= b.x[i];
            if (a.x[i] < 0) a.x[i] += B, a.x[i + 1]--;
        }
        a.x = a.norm(a.x), a.resign();
        return a;
    }
    friend istream &operator>>(istream &is, Bigint &a) {
        string v; is >> v; a = Bigint(v); return is;
    }
    friend ostream &operator<<(ostream &os, Bigint &a) {
        os << a.to_string(); return os;
    }
    friend bool operator<(Bigint a, Bigint b) {
        if (a.sgn != b.sgn) return a.sgn < b.sgn;
        if (a.x.size() != b.x.size())
            return a.x.size() < b.x.size();
        else for (int i = a.x.size() - 1; i >= 0; i--)
            if (a.x[i] != b.x[i]) return a.x[i] < b.x[i];
        return 0;
    }
    friend bool operator>(Bigint a, Bigint b) {
        if (a.sgn != b.sgn) return a.sgn > b.sgn;
        if (a.x.size() != b.x.size())
            return a.x.size() > b.x.size();
        else for (int i = a.x.size() - 1; i >= 0; i--)
            if (a.x[i] != b.x[i]) return a.x[i] > b.x[i];
        return 0;
    }
    friend bool operator==(const Bigint &a, const Bigint &b) {
        return a.sgn == b.sgn && a.x == b.x;
    }
    friend bool operator!=(const Bigint &a, const Bigint &b) {
        return a.sgn != b.sgn || a.x != b.x;
    }
    friend bool operator>=(const Bigint &a, const Bigint &b) {
        return a == b || a > b;
    }
    friend bool operator<=(const Bigint &a, const Bigint &b) {
        return a == b || a < b;
    }
};
Bigint abs(const Bigint &a) { return a.abs(); }
Bigint stoiBigint(const string &s) { return Bigint(s); }

```

11.4 Multiple [7600ae]

```

// Require: Mint, NTT
using i128 = __int128_t;

```

```

vector<i128> arbitraryMult
(const vector<int> &a, const vector<int> &b) {
    auto x = conv(vector<Z1>
        >(a.begin(), a.end()), vector<Z1>(b.begin(), b.end()));
    auto y = conv(vector<Z2>
        >(a.begin(), a.end()), vector<Z2>(b.begin(), b.end()));
    auto z = conv(vector<Z3>
        >(a.begin(), a.end()), vector<Z3>(b.begin(), b.end()));
    vector<i128> res(x.size());
    for (int i = 0; i < x.size(); i++)
        res[i] = garner<i128>(x[i].x, y[i].x, z[i].x);
    return res;
}

public:
friend Bigint operator*(Bigint a, const Bigint &b) {
    a.x = a.norm(arbitraryMult(a.x, b.x));
    a.sgn *= b.sgn, a.resign();
    return a;
}

```

11.5 Division [5be5f1]

```

private:
pair<vector<int>, int> smallDivMod(vector<int> a, int v) {
    ll add = 0;
    for (int i = a.size() - 1; i >= 0; i--) {
        add = add * B + a[i];
        int q = add / v;
        a[i] = q, add %= v;
    }
    return {norm(a), add};
}

friend Bigint operator<<(Bigint a, int k) {
    if (!a.x.empty()) {
        vector<int> add(k, 0);
        a.x.insert(a.x.begin(), add.begin(), add.end());
    }
    return a;
}

friend Bigint operator>>(Bigint a, int k) {
    a.x = vector<int>
        >(a.x.begin() + min(k, int(a.x.size())), a.x.end());
    a.x = a.norm(a.x);
    return a;
}

public:
friend pair<Bigint
, Bigint> divmod(const Bigint &a, const Bigint &b) {
    int qsgn = a.sgn * b.sgn, rsgn = a.sgn;
    Bigint av = a.abs(), bv = b.abs();
    Bigint quot, rem;
    if (av < bv) {
        rem = av;
    } else if (bv.size() == 1) {
        auto [qd, r] = quot.smallDivMod(av.x, bv.x[0]);
        quot.x = qd;
        rem = Bigint((ll)r);
    } else {
        Bigint inv = 1LL * B * B / bv.x.back();
        Bigint pre = 0, res = 0;
        int d = av.size() + 1 - bv.size();
        int cur = 2, bcur = 1;
        while (inv != pre || bcur < bv.size()) {
            bcur = min(bcur << 1, bv.size());
            res.x = {bv.x.end() - bcur, bv.x.end()};
            pre = inv;
            inv = inv
                * ((Bigint(2) << (cur + bcur - 1)) - inv * res);
            cur = min(cur << 1, d);
            inv.x = {inv.x.end() - cur, inv.x.end()};
        }
        inv.x = {inv.x.end() - d, inv.x.end()};
        res = (av * inv) >> av.size();
        Bigint mul = res * bv;
        while (mul + bv <= av) res = res + 1, mul = mul + bv;
        quot.x = quot.norm(res.x);
        rem = av - mul; // 重用 mul, 不再重算一次 quot * b
    }
    quot.sgn = qsgn, quot.resign();
    rem.sgn = rsgn, rem.resign();
    return {quot, rem};
}

friend Bigint operator/(const Bigint
&a, const Bigint &b) { return divmod(a, b).first; }
friend Bigint operator%(const Bigint
&a, const Bigint &b) { return divmod(a, b).second; }

Bigint gcd(Bigint a, Bigint b) {
    while (b != 0) {
        Bigint r = a % b;
        a = b, b = r;
    }
    return a;
}

```

11.6 Division-Python [110bd8]

```

from decimal import * # 無誤差浮點數
setcontext(
    Context(prec=4000000, Emax=4000000, rounding=ROUND_FLOOR))
t = int(input())
for i in range(t):
    a, b = map(Decimal, input().split())
    d, m = divmod(a, b)
    print(d, m)

```